

Background knowledge-

What is an active directory?

An active directory is a collection of tools used to control a computer network

- A computer network includes all users, and their respective computers, connected under the same wifi.
- Routers provide WIFI/Switches for local file sharing/interaction between devices
 - Routers utilize modems to connect to the wider Internet(WAN)
 - Modems translate data from Internet Service Provider (ISP) into data that can be read by devices connected via router
 - The Router transmits the connection from modems to all devices that are connected locally so they can communicate with the internet
 - This includes browsing, streaming, and emailing.
 - They make WAN possible by connecting LAN to another LAN, with a modem
 - WIFI replaced ethernet switches in making a LAN, but still needs a modem.
- Active directory is specifically Microsoft's version and is now industry standard

Active directories store all user information, including their username, password, and permissions. ADs also control security aspects like password expiration dates and password requirements.

- Active directories uphold authentication and authorization

ADs can automatically modify any settings on connected computers and makes it easy to add new users to the network(ex. Think of adding new students to a high school network)

- Essentially, ADs centralized management systems for networks. (specifically for windows)

Domain Controller (DC)

A type of server that provides authentication and authorization for an active directory domain

- It runs the Windows server operating system
- Regular window versions cannot be DCs
- Every AD must have at least one DC
- Each DC stores a copy of the active directory file so any changes made to that file (for example, user changes password) is updated to all active DCs to keep information up to date.
- The active directory file is labeled Ntds.dit
 - Contains details about the users, computers, security groups, and other objects in that domain
 - Includes information about the Active Directory Schema
 - The schema defines all objects that can be stored in the directory

Called a Domain controller because a domain is a group of computers and it controls the groups access to shared network resources and also controls and verifies who logs in.

- Essentially, a DC is a specific type of server. The services it provides include authentication (the user is who they say they are) and authorization (the user has correct permissions)
 - Servers are broad terms for any computer that provides data/services to other computers
 - Ex.
 - File server- stores and manages data files that other computers can access
 - Application server- runs software for client computers
 - Email server- delivers email between clients
 - DNS server- translates human-readable website names into their numerical IP addresses

Domain- Two definitions

Web Domains

- Connects readable name to IP address
- Used to find websites
- When u type website names into browsers Domain Name System (DNS) corresponds the name to the IP, like a phone book
- Includes top-level extensions like .com, .org

Network Domains(Active Directory)

Active directory domain is a type of network domain that shares security policies amongst multiple users AND databases

- The databases are important bc it stores user information
- A group of computers, users, and printers tied together under one shared security database
- Lets any IT worker control all company devices and passwords from one central server. If you log into a work computer using your company ID, the network domain checks your password and gives you access to your files.

NIC- Network Interface Card

Enables data to be sent from computer to other network connected devices

ex . router and switches

NIC acts as an interface between the physical layer(ex. cables) and the data link layer (which is the communication channel).

- NIC converts the signals from cables or antennas into digital signals
 - The data is then sent over ethernet or Wi-Fi waves.
 - NICs are the hardware bridge between a computer and wired or wireless networks

- Converting computer binary into electrical/radio wave signals.
- Assigns and uses a permanent Media Access Control (MAC) address so data reaches the correct device.
 - Uses the MAC address to make sure packets are routed to the correct destination

DHCP- Dynamic Host Configuration Protocol

DHCP helps your device connect to your router by automatically assigning/configuring your device an IP, subnet, mask, gateway, and DNS server.

- Without DHCP you would have to manually enter all IP, subnet, mask, etc.. information yourself when u want to connect
- DHCP helps u connect to LAN networks, you still need a modem to reach the internet
 - Assigns IP address to communicate with devices within the same network

Static IP- IP addresses that never change even if your disconnect and connect to other networks

- You need DHCP to connect dynamically or utilize dynamic ip addresses
- When you connect to a network, DHCP pulls from a pool of IP addresses and assigns your device that address on a lease(a limit on how long that device can use that address)
- Once you disconnect and connect to a new network, it recycles the previous IP addresses and assigns your device a new one from the pool
- This is why it's dynamic, because it changes constantly

DNS

- DNS servers are phonebooks that assign human readable addresses to IP addresses
- Different networks have different DNS servers because they configure them differently(ex. Different settings)

IP(Internet Protocol), Mask, Gateway

IP address- its like your device's address, it tells other devices how to reach your device and where to send data.

- When we type a website name into a browser, the DNS server translates it into the IP address which then tells our device where exactly to send data.

Mask/Subnet Mask-

- Subnet masks are used to differentiate which part of the IP address is the network and which part is the host.
 - Subnet masks tell the computer where to slice the IP address
 - The left part is the network and the right side is the computer on the network (host).
 - The computer can only tell by using a subnet

- Subnet masks can tell a computer if it needs to send packets to a device on a local network or if it needs to send it to a remote network.
 - If the packet needs to go to a remote network, it needs to go through the gateway/router
 - Also the router is largely considered the default gateway because it is the entry point between locally connected devices and the internet
 - Remember that the router transmits modem signals to all devices
 - When communicating locally, the subnet mask tells the device to NOT send packets to the router and instead the device sends the packet directly to the other device on the LAN.
 - Gateways are all-in-one access points that combine the functions of modems and routers
 - Makes the use of separate routers and modems obsolete.

Gateway- a piece of networking hardware or software that allows data to flow from one network to another

RAS/NAT

RAS- Remote Access Service

NAT- Network Address Translation

- Managed together with Microsoft's Routing and Remote Access Service (RRAS)

RAS

- Allows individual users or remote sites to connect surely to a private corporate network from the outside (typically uses a VPN to do this)
 - Authenticates users, assigns remote IP, and manages secure session entries

NAT

- Translates private IP addresses into a single public IP address when the device communicates with external internet resources(ex. browsing)
 - Saves IPv4 addresses and hides internal network structures from outside view

How they work together

Combined routing

- Operating system routing tools often pair RAS and NAT on a gateway server(a DC) with one interface(one NIC) facing the private network and the other (one NIC) facing the public internet.

Shared Traffic Flow

- RAS handles incoming remote user authentication, NAT translates packet addresses so connected users or LAN devices can reach the external internet through the shared gateway.

Project Walkthrough

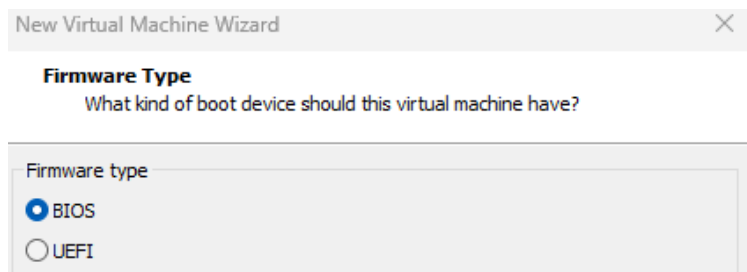
Requirements:

2 Virtual Machines

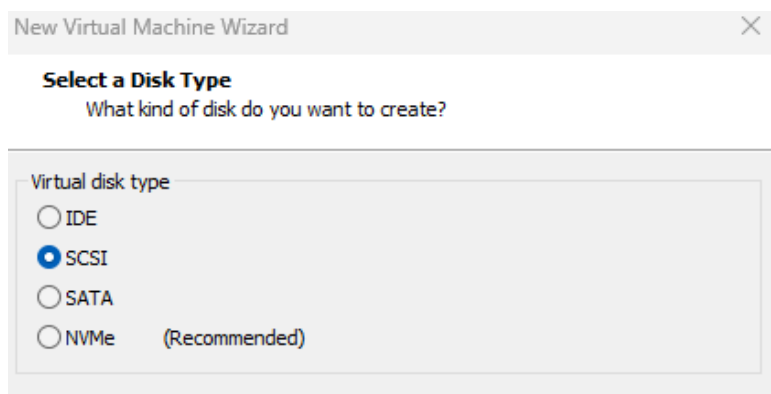
Windows Server 2019 Virtual Machine

- This VM acts like the domain controller
- 2048mb(2GB) of RAM
- 2 CPU Cores
- 20GB Disk Size
- Username- abhik Password- password

Step 1: Configuring Domain Controller Virtual Machine



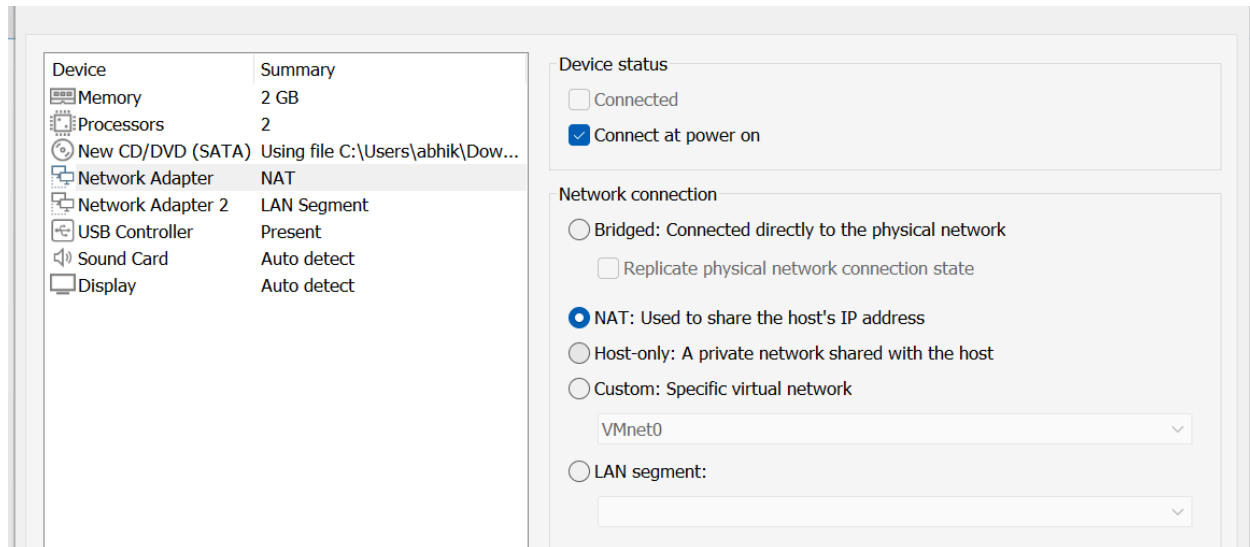
When configuring the VM I chose BIOS as the firmware type.



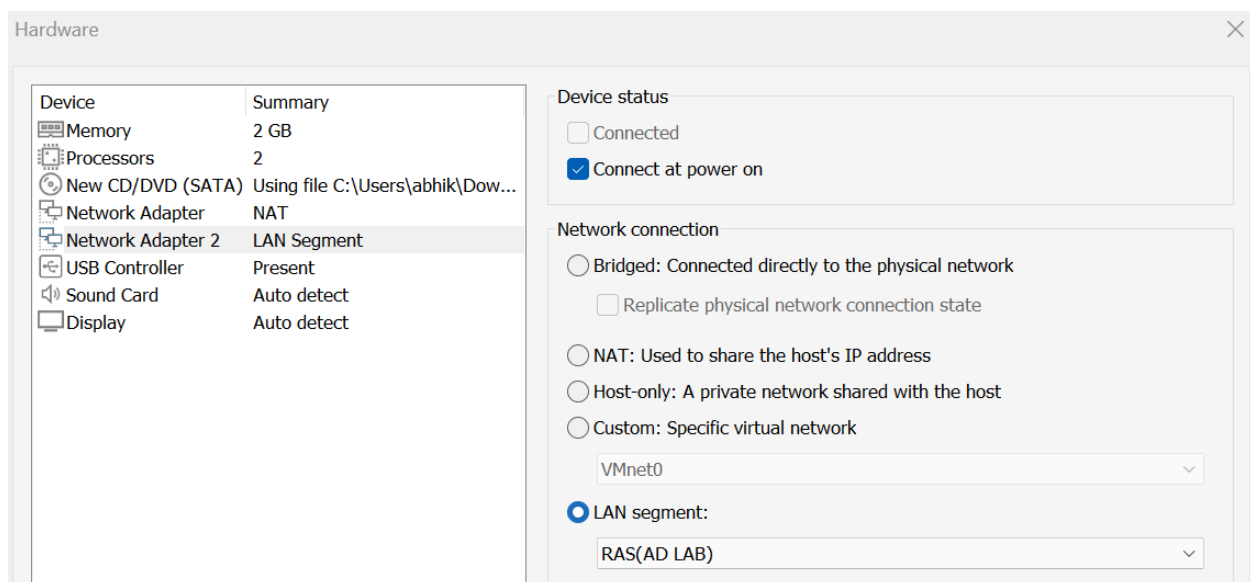
I chose SCSI as my disk type.

Step 2: Configuring Network Settings on DC VM

NAT and Internal Network(LAN) (two NICs)

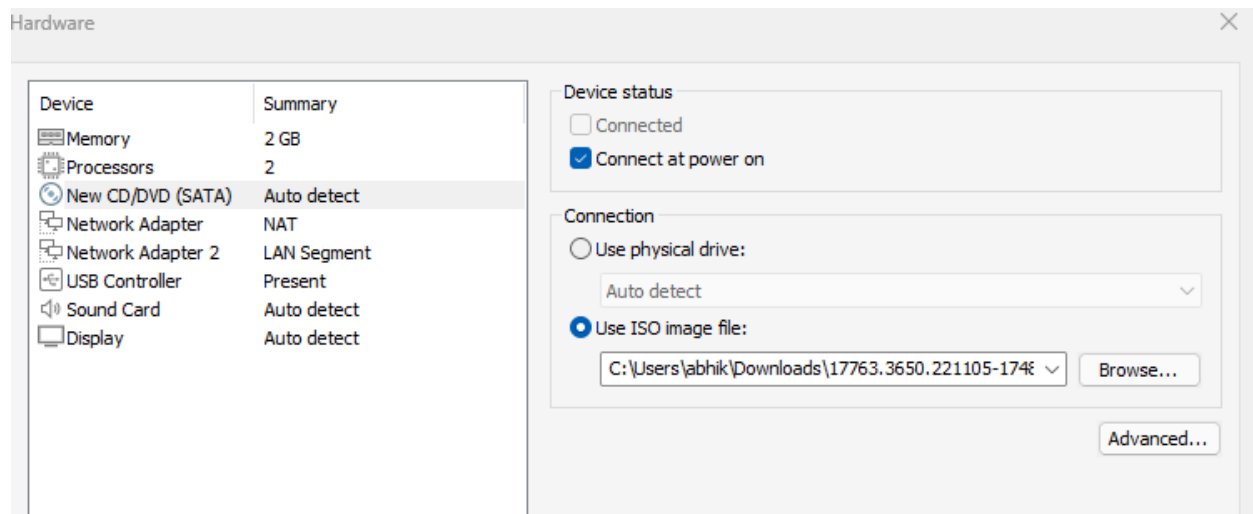


One of our network adapters needs to be a NAT, Network Address Translator, because we still need to be able to access the external internet from our LAN.



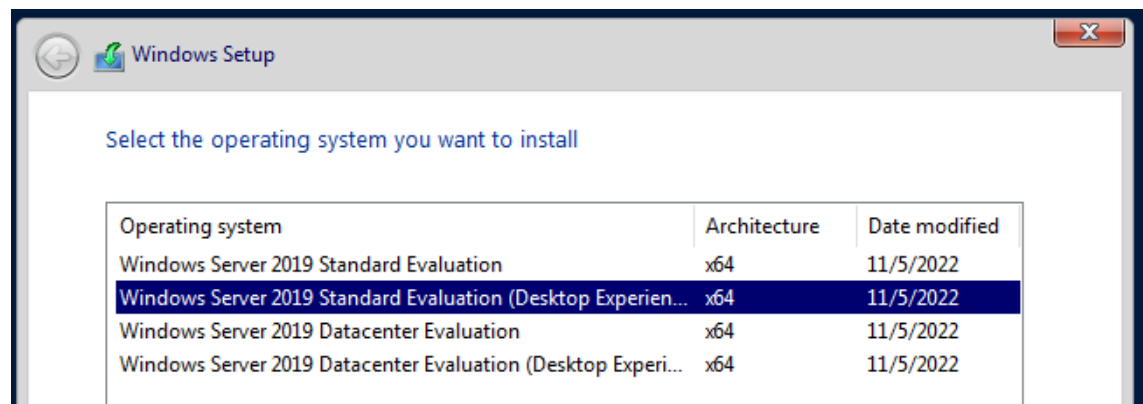
Our internal network(LAN) adapter allows our client VM to communicate with other devices on the same network securely.

- The purpose of an internal network in a functioning enterprise environment is to keep data shared between devices on the same network secure and prevent the network infrastructure from being revealed to outsiders.

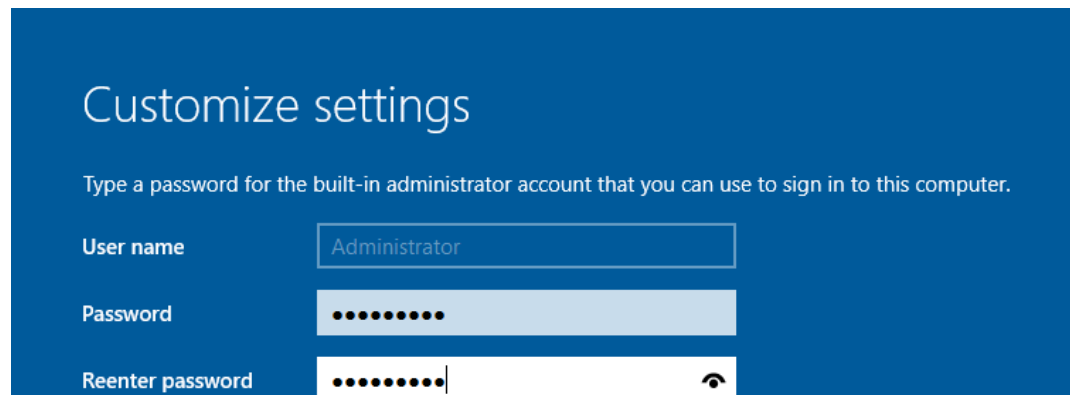


Mounted the Windows Server 2019 iso file to the VM.

I proceeded with the rest of the Windows Server setup by selecting English as the interface and keyboard language and instructed my VM to install all related files to the 20GB disk space we allocated earlier.



Selected the Standard Evaluation (Desktop Experience) so the server installs with a GUI

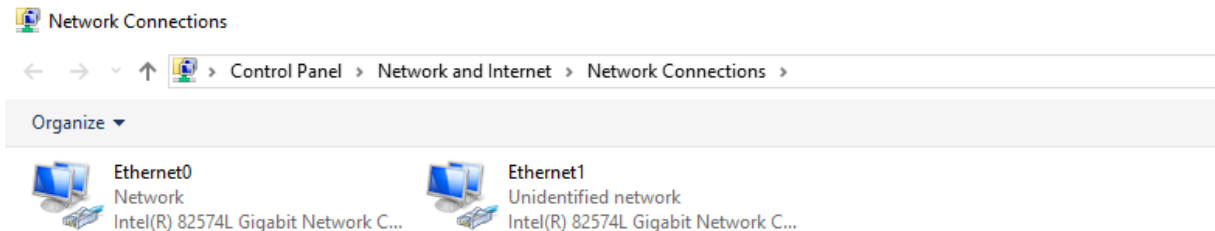


Created the Administrator Account

Step 3: Setting up IP Addressing(External and Internal(LAN) connections)

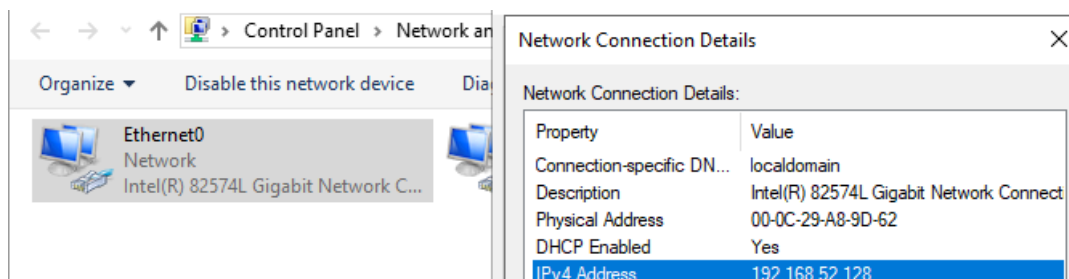
External NIC gets its IP address from my home router, or inherits the IP address of my host computer

The Internal NIC, the one responsible for our LAN network, is the one we need to configure.

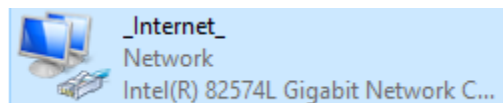


We need to go to ethernet settings and click on change adapter options

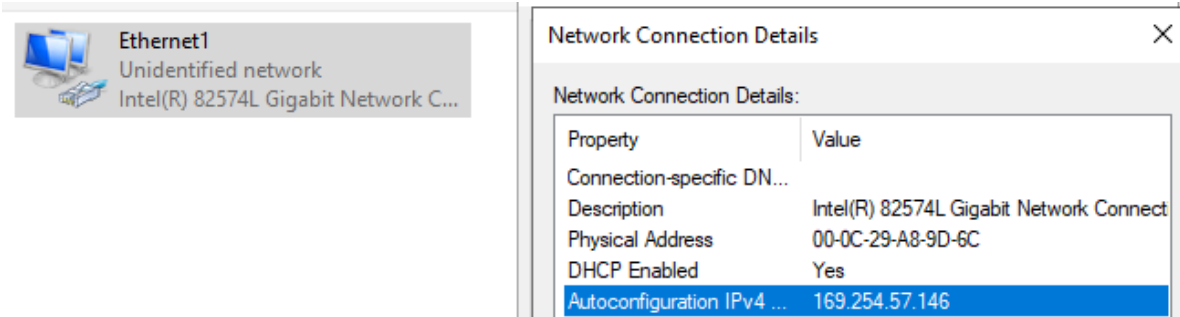
- Remember when we first configured the VM, we created two network adapters.



- Ethernet0 is our NAT adapter, or the adapter responsible for communication with the external internet
 - We could tell because the IP address assigned to it is within the typical 192.168 range, which is reserved for private home or office networks.

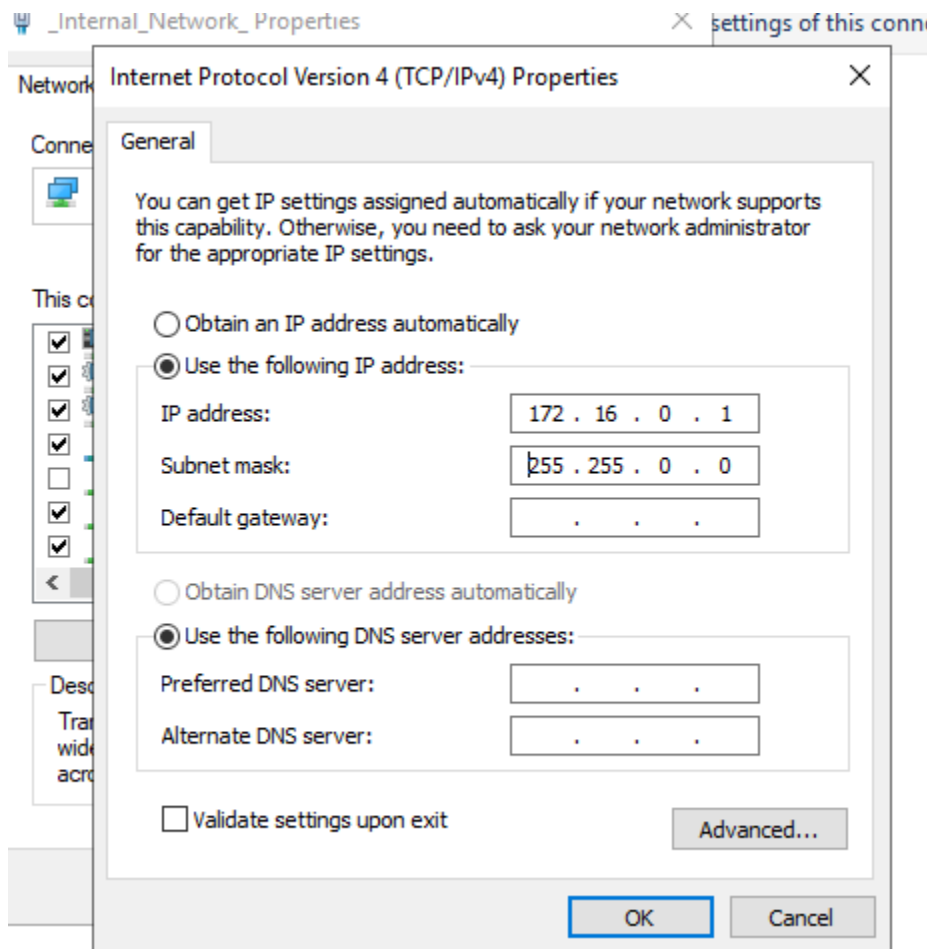


Renamed "Ethernet0" to "_Internet_" for clarity.

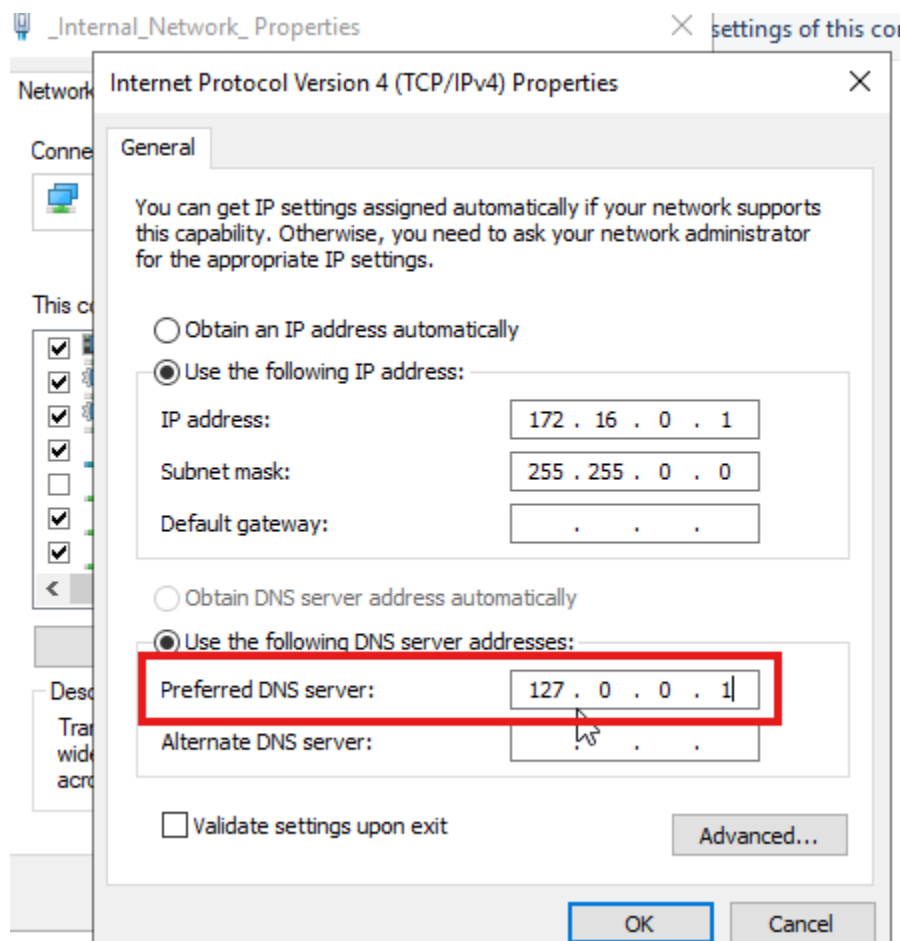


- You can tell Ethernet1 is our LAN network adapter because when an adapter is assigned the IP address 169.254.xx.xx it means the adapter couldn't find a DHCP server and automatically assigned an IP address.
 - Renamed Ethernet1 to "_Internal_Network_" for clarity.

Step 4: Assigning IP Address for LAN network

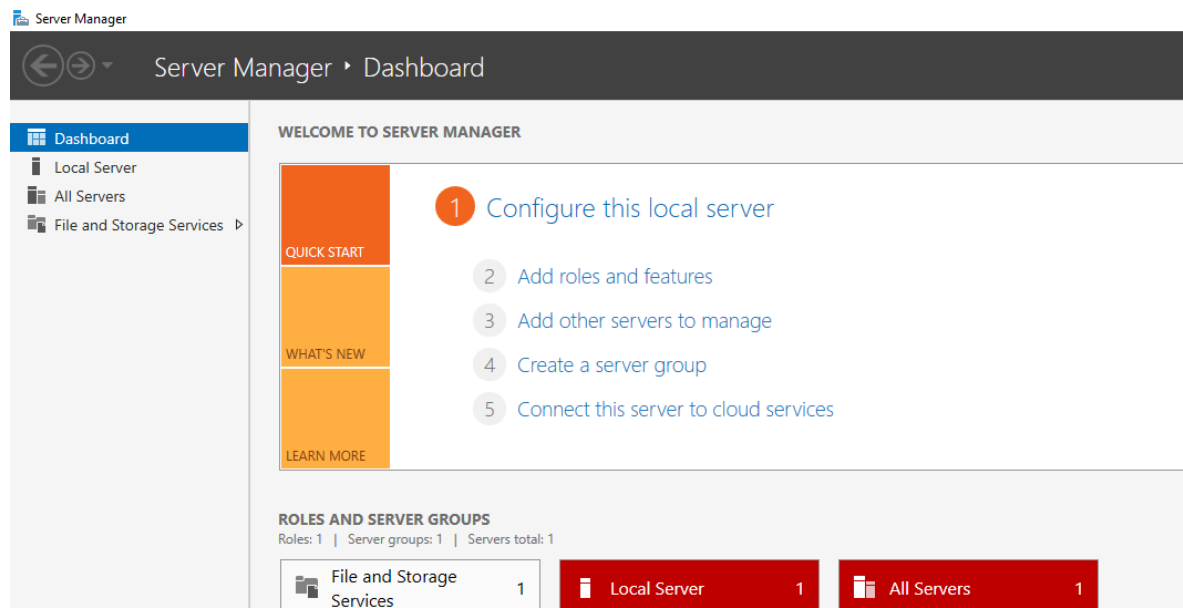


- I edited the IPv4 properties of the LAN adapter and I didn't add a specific Default Gateway because our Domain Controller itself will function as the default gateway for our internal network.



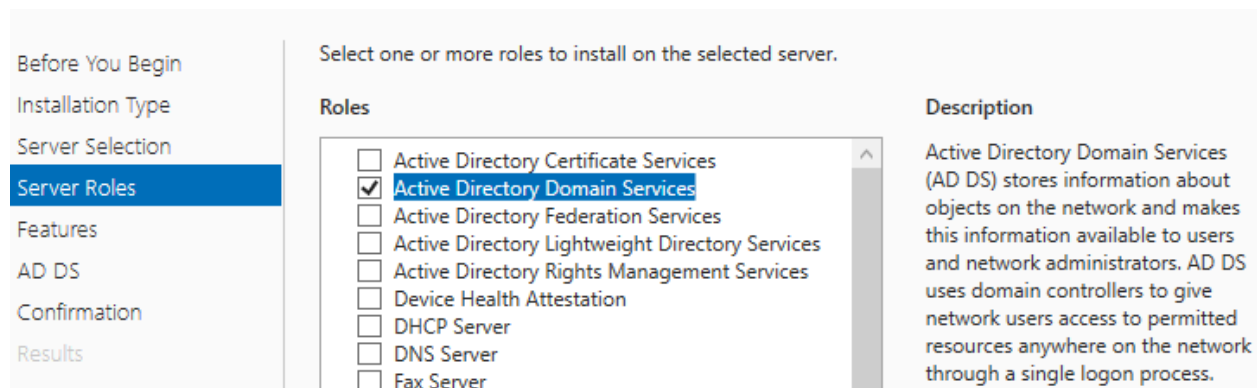
- When we install Active Directory/Promote a server to domain controller, DNS is automatically installed. Essentially, by entering 127.0.0.1 as the preferred DNS server, we are telling the adapter on this server to refer back to itself for a DNS.
 - 127.0.0.1 is the IP address that tells the server to ping themselves, you can also use the IP address of the server as the preferred DNS server (172.16.0.1) since that also tells the server to refer back to itself.

Step 5: Installing Active Directory Domain Services (AD DS) and creating a domain



To install Active Directory Services we need to go to the Server Manager and click on “Add roles and features”.

- We are choosing role-based installation because we want the standard installation path.



We are choosing Active Directory Domain Services as the server role.

- We can see in the description given, installing this role allows us to make information about users available to network administrators and utilize domain controllers to give network access to verified users.

Confirm installation selections

DESTINATION SERVER
DC

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

Confirmation

Results

To install the following roles, role services, or features on selected server, click Install.

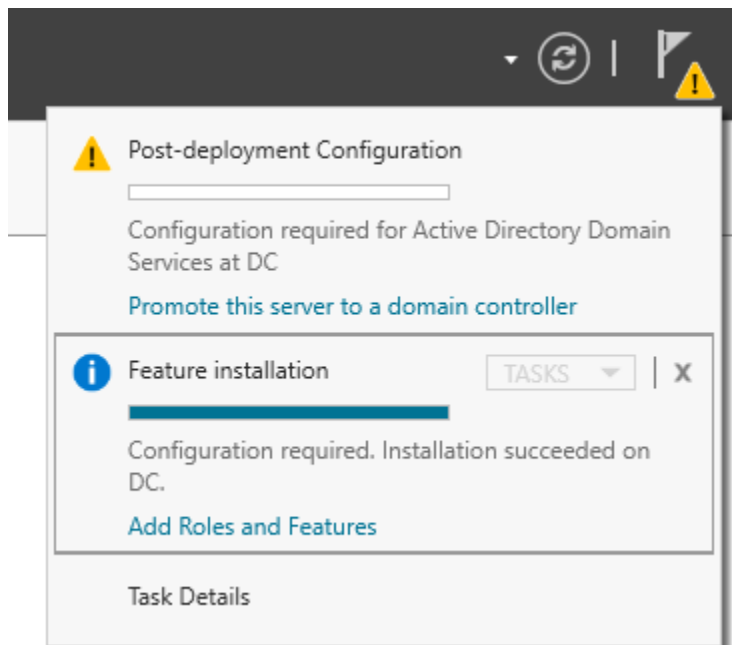
☐ Restart the destination server automatically if required

Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.

Active Directory Domain Services
Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell
AD DS Tools
Active Directory Administrative Center
AD DS Snap-Ins and Command-Line Tools

- All of these services and roles will be installed on our DC.

Step 6: Promoting our server to a Domain Controller



- Since we installed new services and roles on our server, we can now officially promote our server to a domain controller.

Deployment Configuration

TARGET SERVER
DC

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

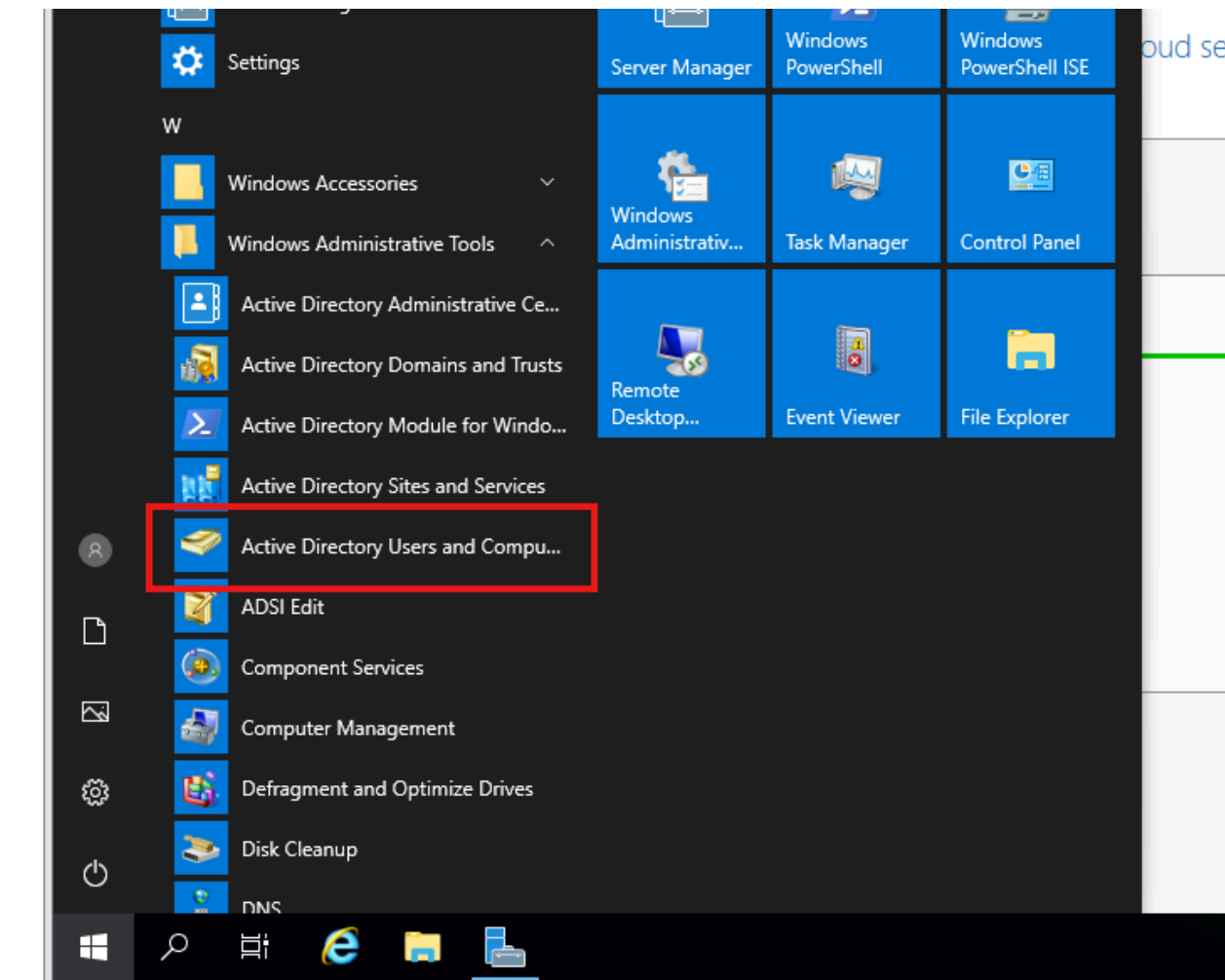
< Previous Next > Install Cancel

- We are adding this server to a new forest. Forests are simply the containers for our Active Directory. They allow for shared rules, all users must follow the same rule, and act as a security barrier preventing users within this forest from accessing other forests
 - We also named our domain mydomain.com
 - This will become important once we start configuring our client VM
 - Simply click next for all other tabs until the install option is available
 - Once we finish configuring our domain, our computer will restart

Step 6: Creating our own Domain Admin Account

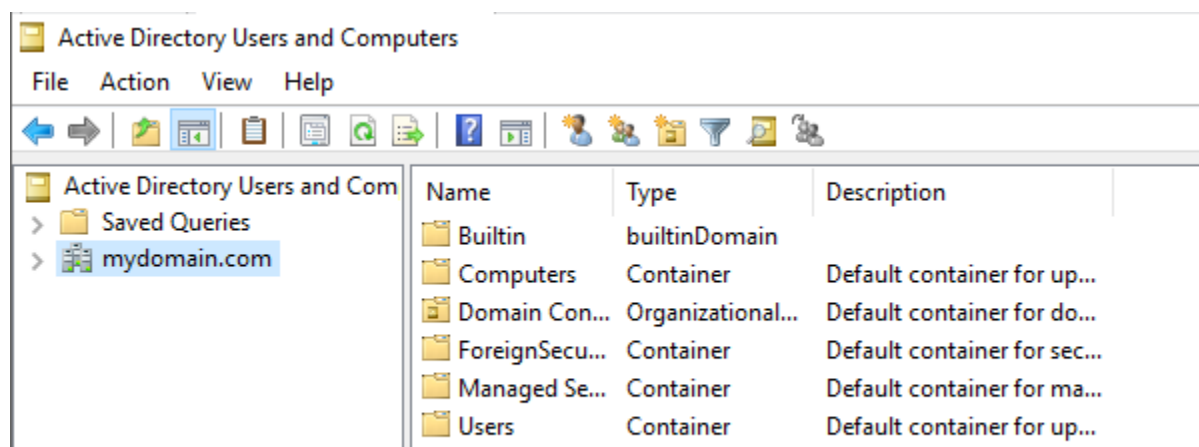
We want to create our own Domain Admin Account so we can be the user with the highest permissions

- This will become useful when we start to organize our users that we generated from the PowerShell script into different groups with different permissions.
- This is how we can practice the concept of least privilege which is important in enterprise networks and is a key part of both SOC work and GRC work.

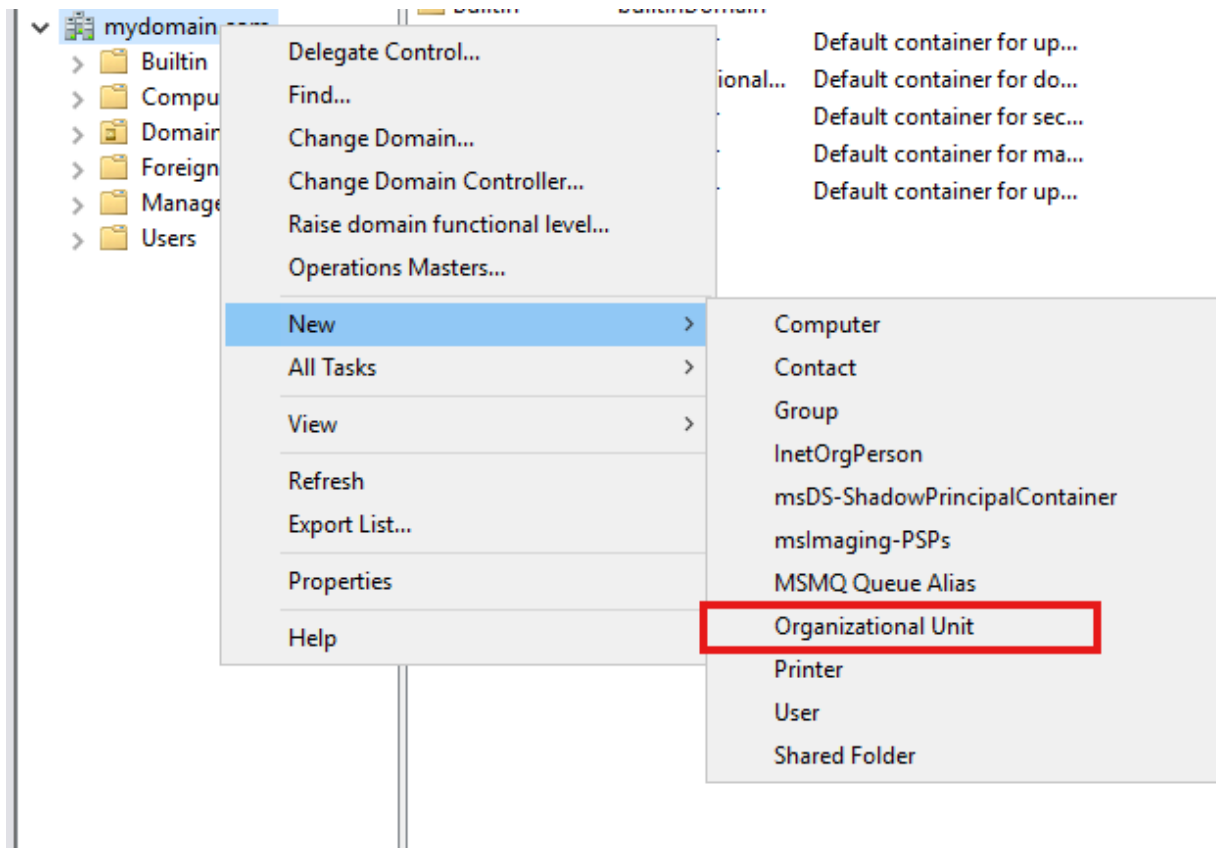


Once our computer restarts, we should navigate to the start menu and click on “Active Directory Users and Computers”.

- This is where we can see all the users in our Active Directory and all connected computers.

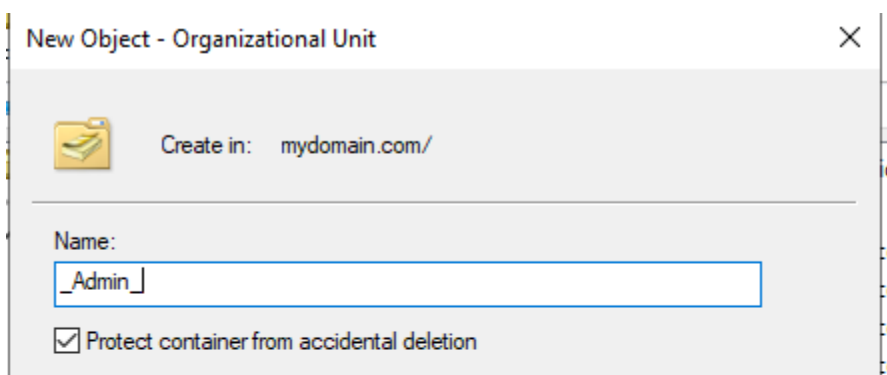


- We can see the Active Directory we created here.

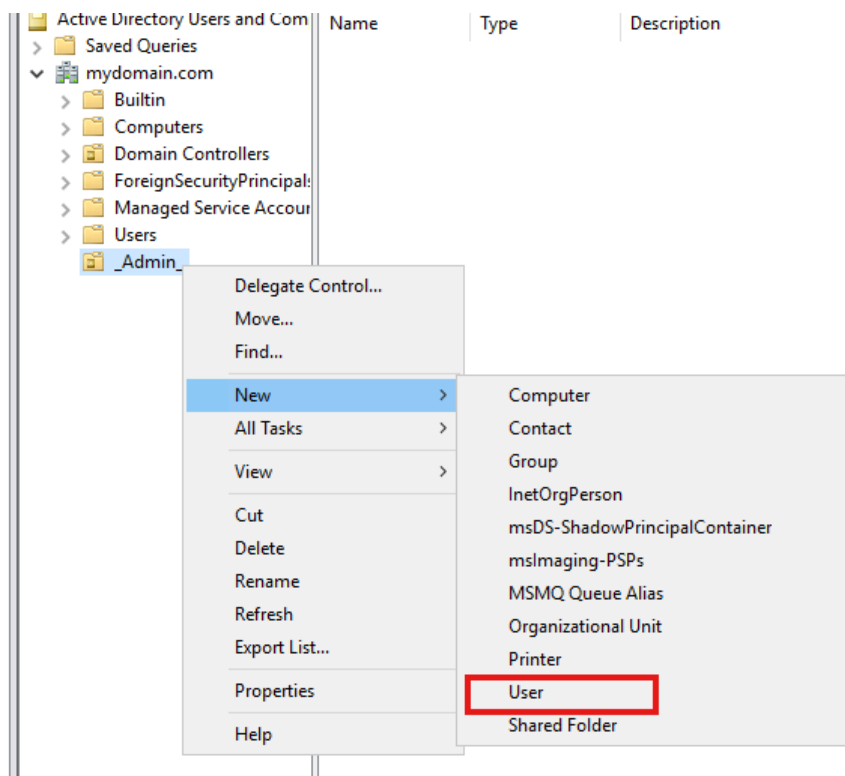


Right click, click new, and then click organizational units.

- We want to create an organization unit that houses only admin accounts.
 - This will make it easier to configure permissions for all users who are considered admins and implement least privilege.



- You should give your organization unit a descriptive name so a user/administrator knows what type of users each unit contains.



We can see our new organization unit under mydomain.com.

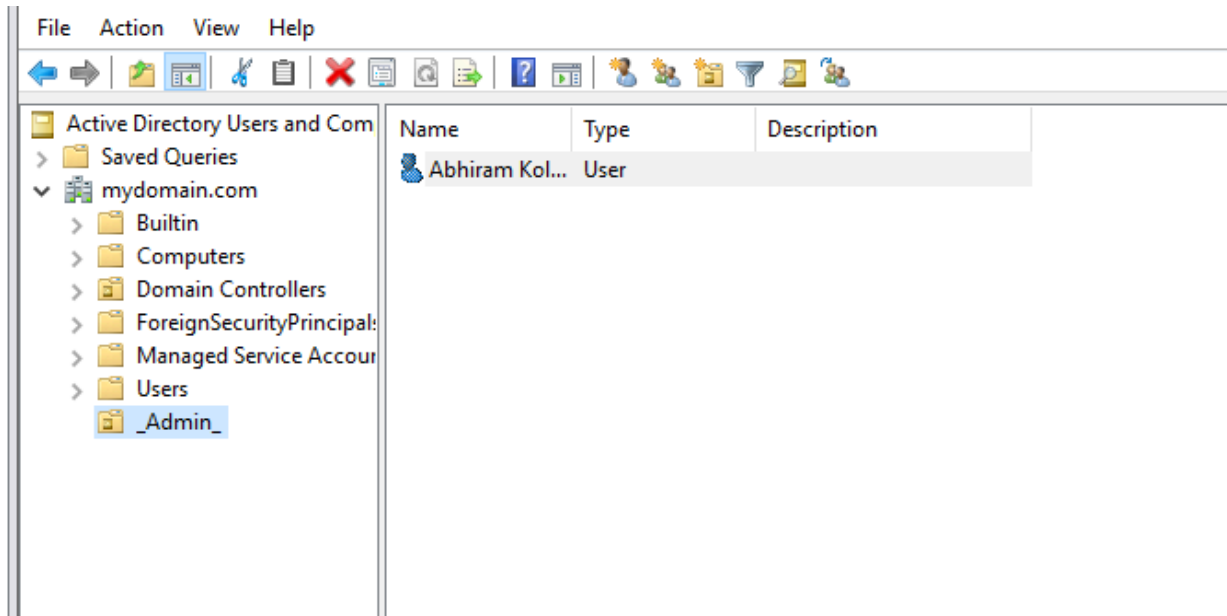
- Right click. Click new, and then click user to make a new user for our admin unit.

The screenshot shows the 'New Object - User' dialog box. The 'Create in:' field is set to 'mydomain.com/_Admin_'. The 'First name' field contains 'Abhiram', the 'Last name' field contains 'Kolanu', and the 'Full name' field contains 'Abhiram Kolanu'. The 'User logon name' field contains 'a-abhik' and the domain dropdown is set to '@mydomain.com'. The 'User logon name (pre-Windows 2000)' field contains 'MYDOMAIN\' and the 'a-abhik' field. The 'Next >' button is highlighted.

Name	Type	Description
Abhiram		
Kolanu		
Abhiram Kolanu		
a-abhik		
MYDOMAIN\'		
a-abhik		

Fill out the information for the new user.

- a-(username) is a common naming convention in many organizations when the user is considered an admin.
- Click next and you will enter a password for the user.



- We created our user in the admin organization unit, but now we need to give admin permissions.

Abhiram Kolanu Properties

Member Of	Dial-in	Environment	Sessions
Remote control	Remote Desktop Services Profile	COM+	

General Address Account Profile Telephones Organization

Abhiram Kolanu

First name: Initials:

Last name:

Display name:

Description:

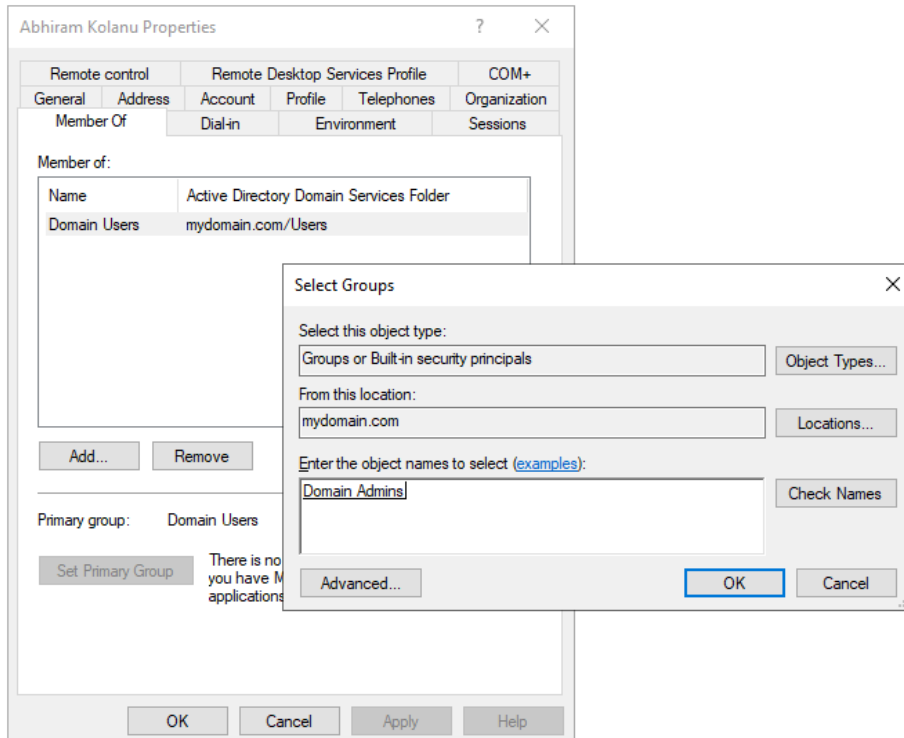
Office:

Telephone number: Other...

E-mail:

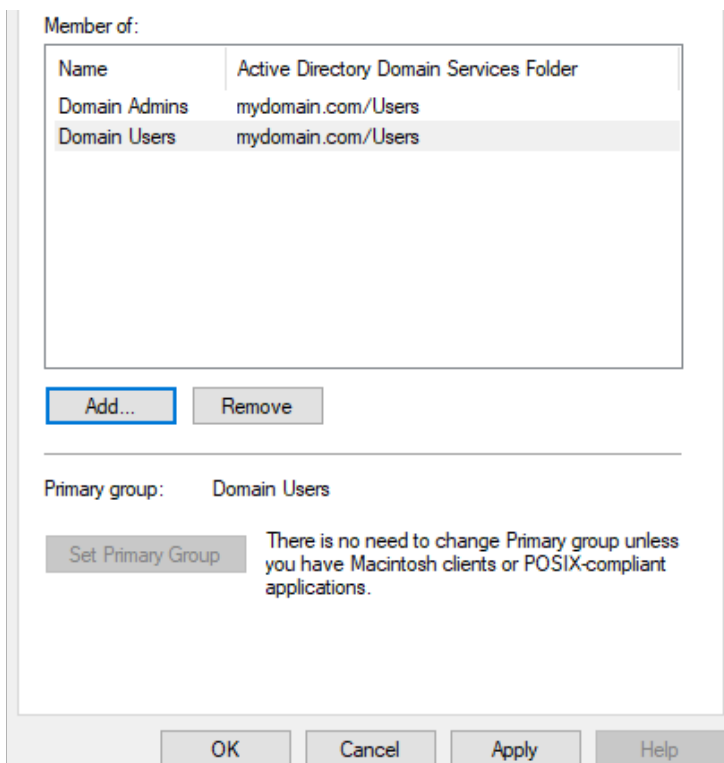
Web page: Other...

If you right click a go to properties, you should get this screen.



Navigate to Member Of, click on add, and then enter “domain admins”

- After you enter “domain admins”, you should click check names and it will underline itself.

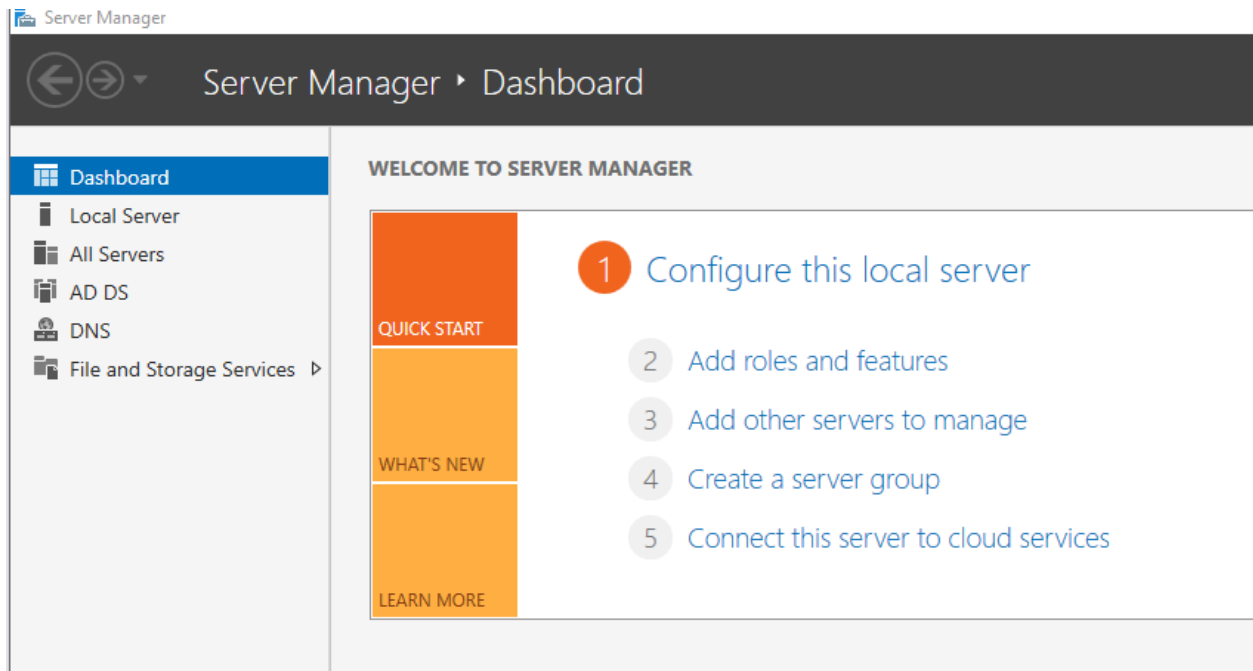


Now apply the new properties.

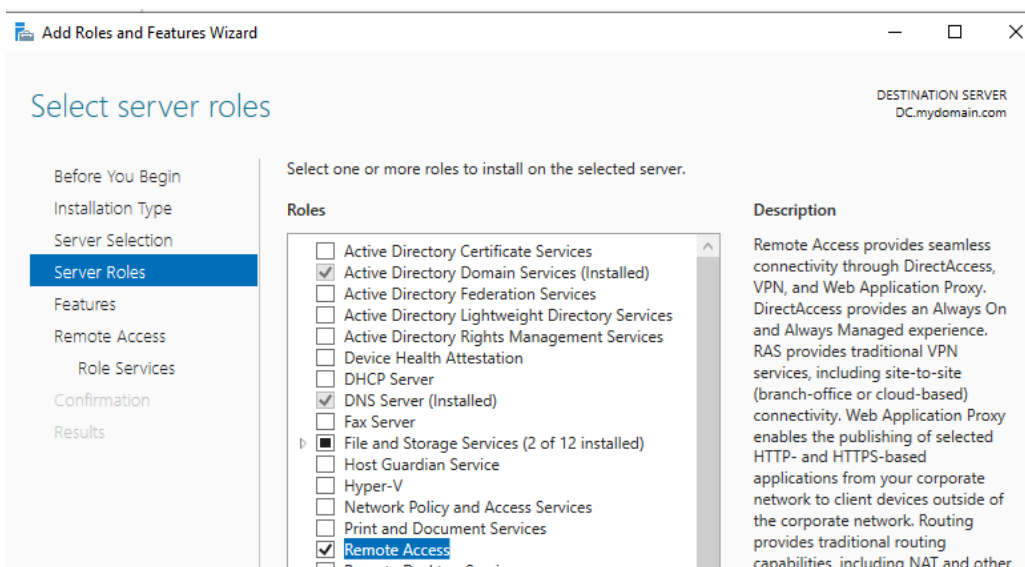
You can now sign out of your machine and login with your new account and operate with admin level privileges.

Step 7: Install RRAS/RAS and NAT

- The purpose of installing these two network technologies is that RRAS allows our client machine, once we install it, to operate in a remote environment while still being able to browse the external internet using NAT(Network Address Translation)



- To begin the installation, we must navigate back to the server manager and click on “add roles and features”



- Click next until “Server Roles” and check “Remote Access”

Select role services

DESTINATION SERVER
DC.mydomain.com

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Remote Access

Role Services

Select the role services to install for Remote Access

Role services	Description
☒ DirectAccess and VPN (RAS)	
☒ Routing	Routing provides support for NAT Routers, LAN Routers running BGP, RIP, and multicast capable routers (IGMP Proxy).
☐ Web Application Proxy	

- We want to select routing here because it will provide support for NAT Routers and LAN Routers.

Confirm installation selections

DESTINATION SERVER
DC.mydomain.com

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Remote Access

Role Services

Web Server Role (IIS)

Role Services

Confirmation

Results

To install the following roles, role services, or features on selected server, click Install.

☐ Restart the destination server automatically if required

Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.

RAS Connection Manager Administration Kit (CMAC)

Remote Access

- DirectAccess and VPN (RAS)
- Routing

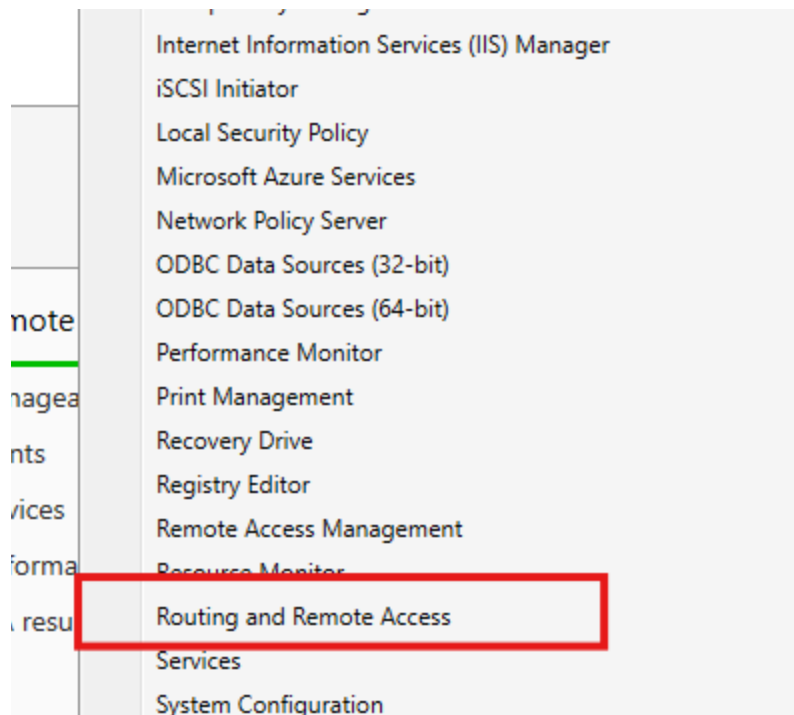
Remote Server Administration Tools

Role Administration Tools

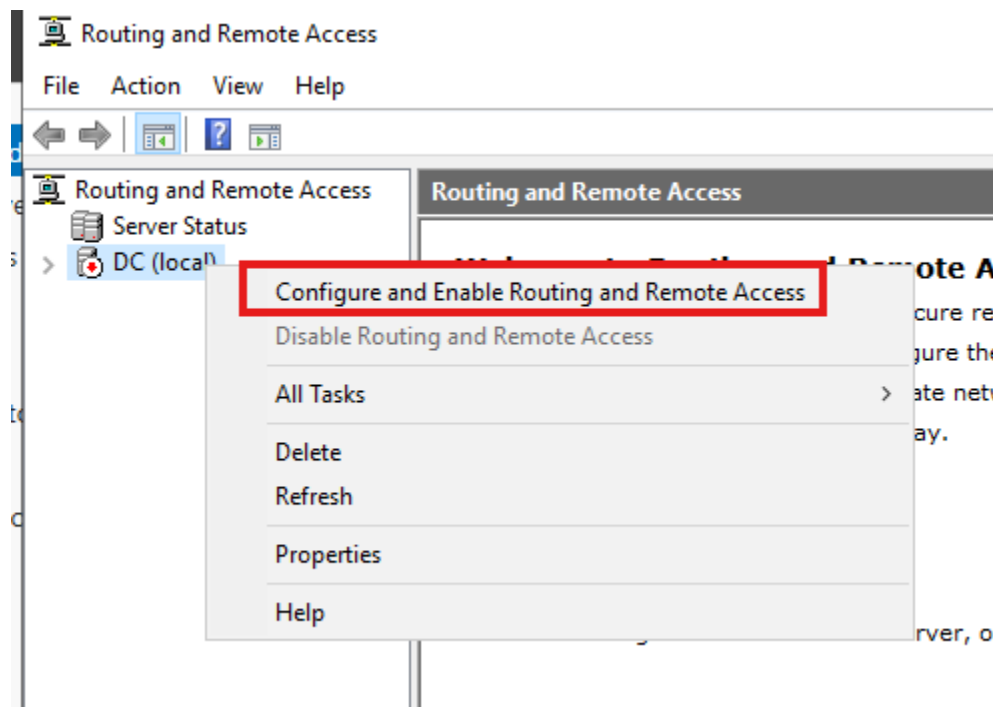
- Remote Access Management Tools
 - Remote Access GUI and Command-Line Tools
 - Remote Access module for Windows PowerShell

Web Server (IIS)

Keep clicking next until you reach this tab and click install.



- After installation, navigate to the tools tab, on the Server Manager, and select routing and remote access



Right click on "DC (local)" and click on "Configure and Enable Routing and Remote Access".

Configuration

You can enable any of the following combinations of services, or you can customize this server.

- ☐ Remote access (dial-up or VPN)
Allow remote clients to connect to this server through either a dial-up connection or a secure virtual private network (VPN) Internet connection.
- ☒ Network address translation (NAT)
Allow internal clients to connect to the Internet using one public IP address.
- ☐ Virtual private network (VPN) access and NAT
Allow remote clients to connect to this server through the Internet and local clients to connect to the Internet using a single public IP address.
- ☐ Secure connection between two private networks
Connect this network to a remote network, such as a branch office.
- ☐ Custom configuration
Select any combination of the features available in Routing and Remote Access.

< Back

Next >

Cancel

Select NAT since that is what we need in order to browse the internet from our secure remote enterprise network.

Routing and Remote Access Server Setup Wizard

NAT Internet Connection

You can select an existing interface or create a new demand-dial interface for client computers to connect to the Internet.

- ☒ Use this public interface to connect to the Internet:

Network Interfaces:

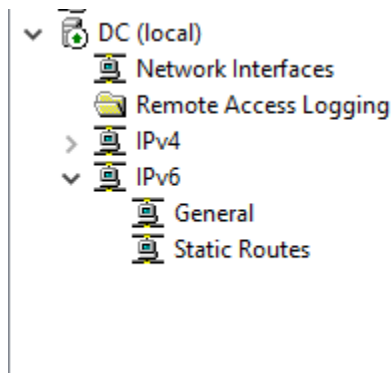
Name	Description	IP Address
_Internal_Network_	Intel(R) 82574L Gigabit...	172.16.0.1
Internet	Intel(R) 82574L Gigabit...	192.168.52.128 (DHCP)

- ☐ Create a new demand-dial interface to the Internet

A demand-dial interface is activated when a client uses the Internet. Select this option if this server connects with a modem or by using the Point-to-Point Protocol over Ethernet. The Demand-Dial Interface Wizard will start at the end of this wizard.

At the beginning of the lab, I renamed my adapters and I will select the one I renamed “_Internet_” because that is the adapter I want to use to connect my VMs to the internet.

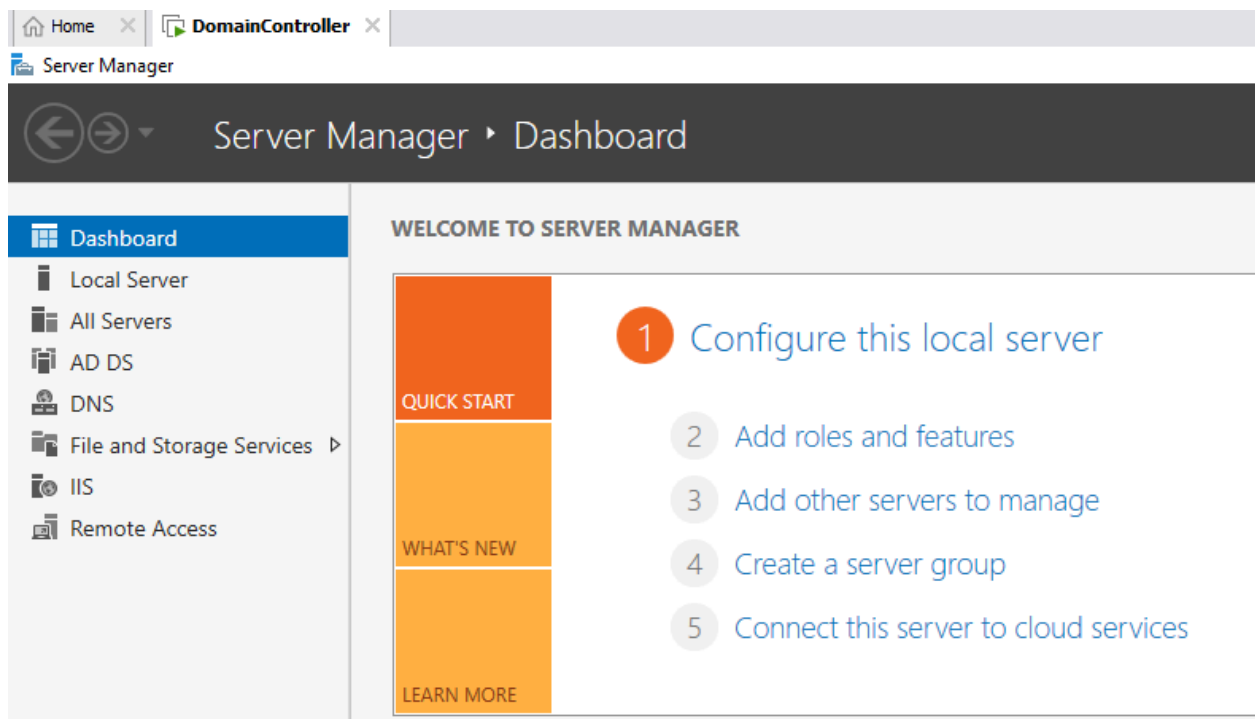
If you remember correctly, the “_Interne_t” adapter gets its DHCP scope from my home router, so no configurations are needed from my side.



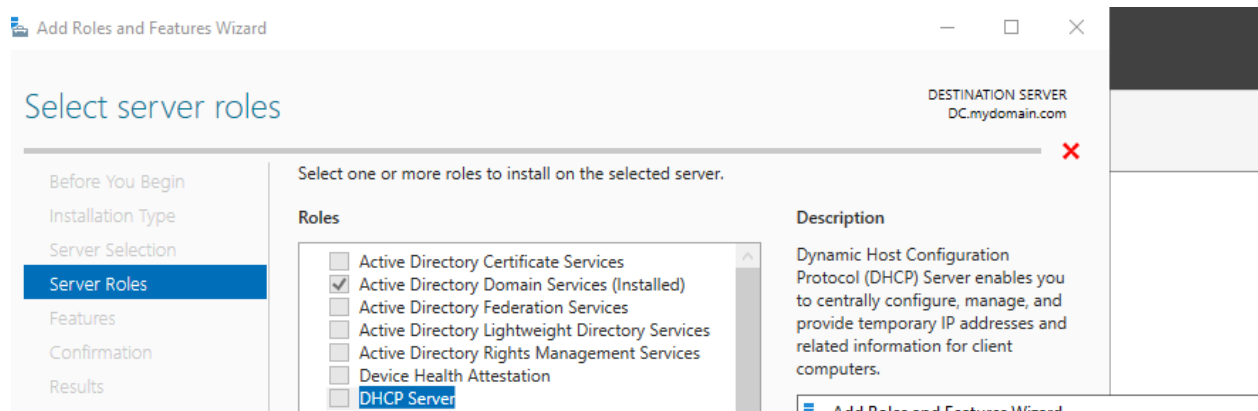
- In the first screenshot, the arrow on the icon was red and pointing downwards, now the arrow is green and pointing upwards.
 - This means our configurations were successful and we have enabled routing and remote access on our server.

Step 8: Setting up a DHCP Server on our Domain Controller

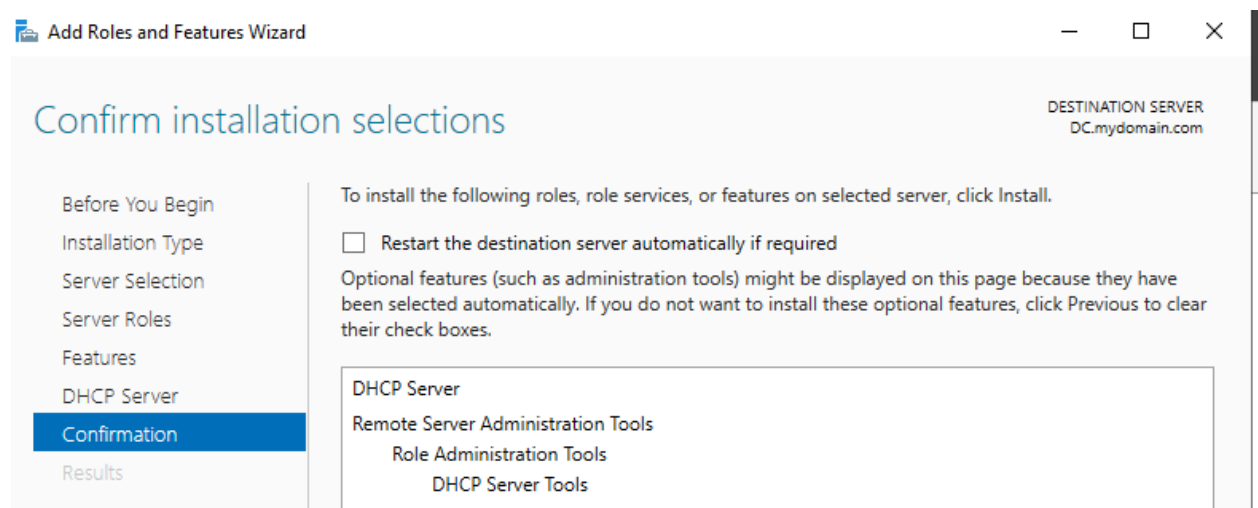
- We want to set up a DHCP server on our Domain Controller so when our Windows 10 Client VM wants to access the internet, we can assign it a public IP address and it can browse like normal.
- Our configuration will include an IP range, gateway, mask, DNS, and lease.



Go to the server manager and click on “add roles and features”

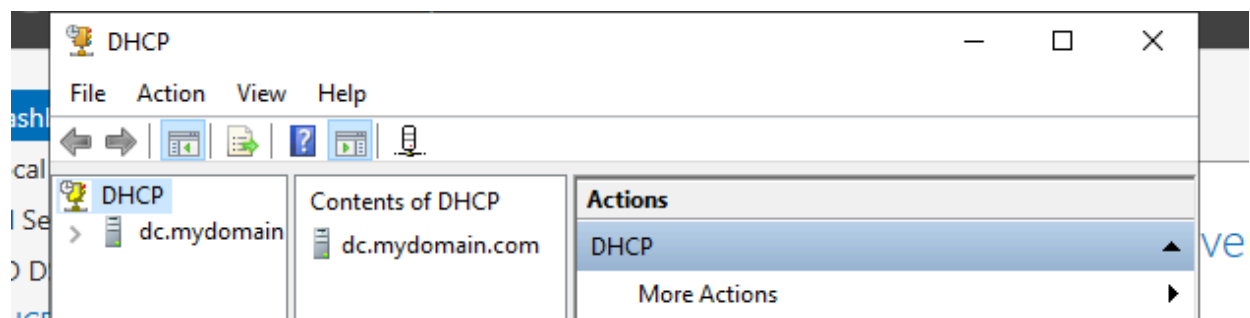


Click next until “Server Roles” and then select “DHCP Server”.



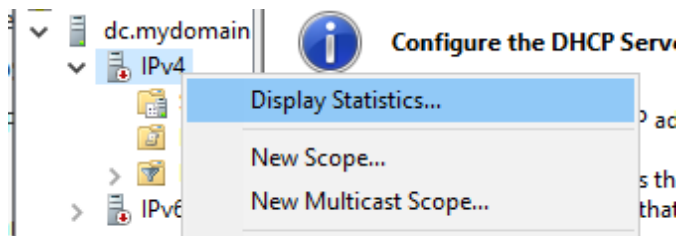
Click next until “Confirmation” and click install.

- After installation is complete, click close.



I navigated to tools, on server manager, and clicked on DHCP so we can begin to configure our scope.

- The core function of DHCP is to allow client computers on the network to receive their IP addresses automatically.



Navigate to IPv4 and click on “new scope”.

- Select a name for the scope. A common naming convention is naming the scope the range of IP addresses you intend to use it for.

New Scope Wizard

IP Address Range

You define the scope address range by identifying a set of consecutive IP addresses.



Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 172 . 16 . 0 . 100

End IP address: 172 . 16 . 0 . 200

Configuration settings that propagate to DHCP Client

Length: 24

Subnet mask: 255 . 255 . 255 . 0

< Back Next > Cancel

- Enter the IP address range and the length
- A mask of 255.255.255.0 will split the IP address right down the middle.
- We can skip “Add exclusions and delay” since this is a contained lab environment and there are no threats yet.
- Lease duration is simply how long a computer “claims” a certain IP address
 - It is how long a computer can have a certain IP address before being refreshed
 - For example, a cafe might set their lease to 2 or 3 hours since they will constantly have new customers throughout the day and want to make space for new connections after previous customers leave.
 - Since this is a lab environment, I left it at 8 days.

New Scope Wizard

Configure DHCP Options

You have to configure the most common DHCP options before clients can use the scope.

When clients obtain an address, they are given DHCP options such as the IP addresses of routers (default gateways), DNS servers, and WINS settings for that scope.

The settings you select here are for this scope and override settings configured in the Server Options folder for this server.

Do you want to configure the DHCP options for this scope now?

☒ Yes, I want to configure these options now

☐ No, I will configure these options later

< Back **Next >** Cancel

Select yes because we do want to configure these options.

- We want to tell the clients what servers to use for DNS and what server to use for a gateway.

New Scope Wizard

Router (Default Gateway)

You can specify the routers, or default gateways, to be distributed by this scope.

To add an IP address for a router used by clients, enter the address below.

IP address:

. . .

Add

Remove

Up

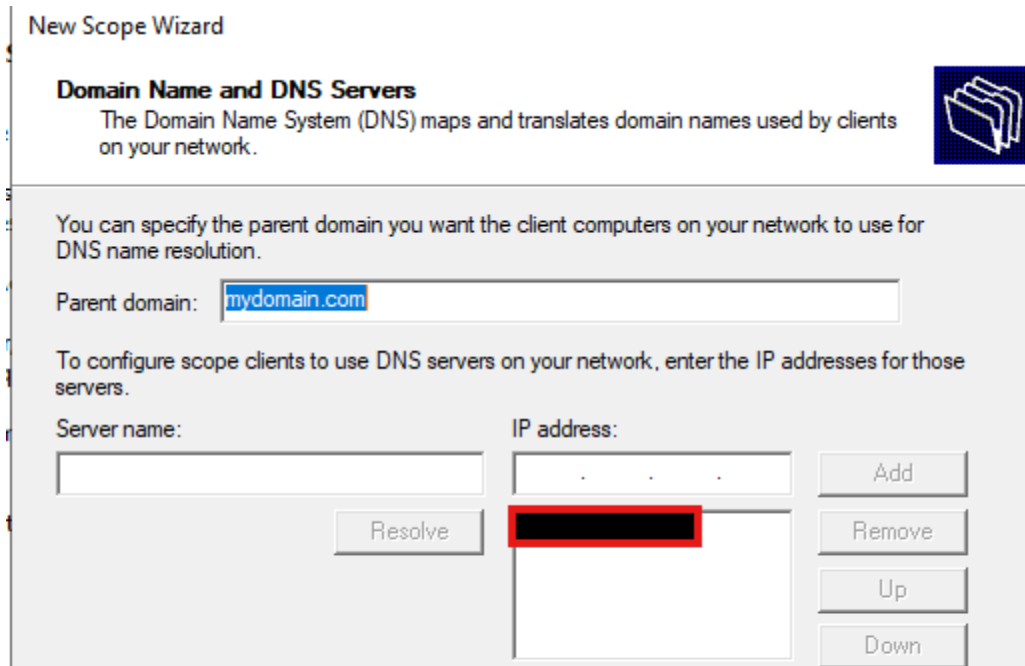
Down

We have already configured our Domain Controller to act like a default gateway.

- We have configured NAT on our Domain Controller, so one of its responsibilities is to forward traffic from the client to the internet, which is a core function of a gateway.

We also enabled routing on this server, which lets it pass data packets across networks(ex. From a local network to the internet).

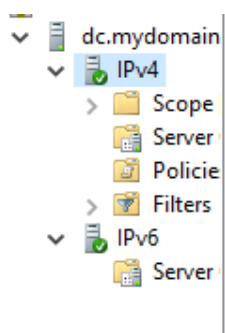
- Since the DC is the gateway itself, simply enter the IP address of the DC as the IP address of the gateway.
 - Make sure to click "Add".



The screenshot shows the 'New Scope Wizard' window, specifically the 'Domain Name and DNS Servers' step. The window has a title bar 'New Scope Wizard' and a sub-header 'Domain Name and DNS Servers'. Below the sub-header is a description: 'The Domain Name System (DNS) maps and translates domain names used by clients on your network.' There is a blue folder icon to the right. The main content area says: 'You can specify the parent domain you want the client computers on your network to use for DNS name resolution.' Below this is a text box labeled 'Parent domain:' containing 'mydomain.com'. Another instruction says: 'To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.' Below this are two columns: 'Server name:' with an empty text box, and 'IP address:' with a text box containing three dots. To the right of the 'IP address:' box is a red rectangular box. Below the 'IP address:' box is a list box with a red border. To the right of the list box are four buttons: 'Add', 'Remove', 'Up', and 'Down'. A 'Resolve' button is located below the 'Server name:' box.

When you install Active Directory on a server, DNS is one of the many roles that are installed.

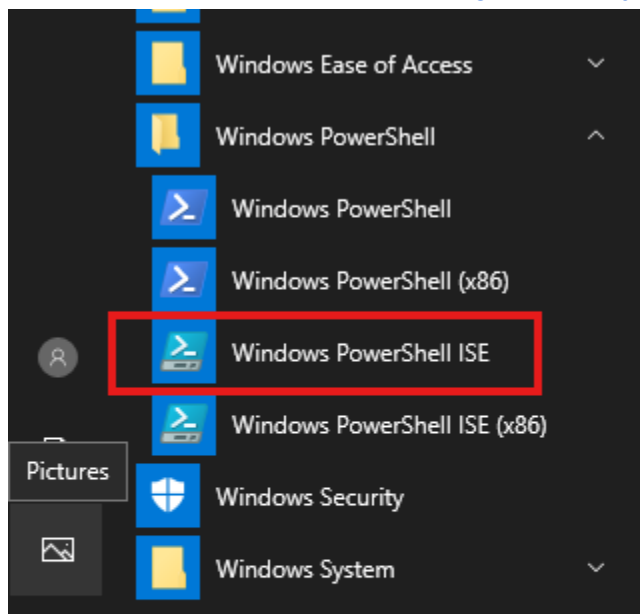
- Since DNS functions are already installed onto our DC, we can use our DC as a DNS as well.
- Enter the IP address of your DNS server and click add.
- Make sure to select "Yes, I want to activate this scope now" at the end.



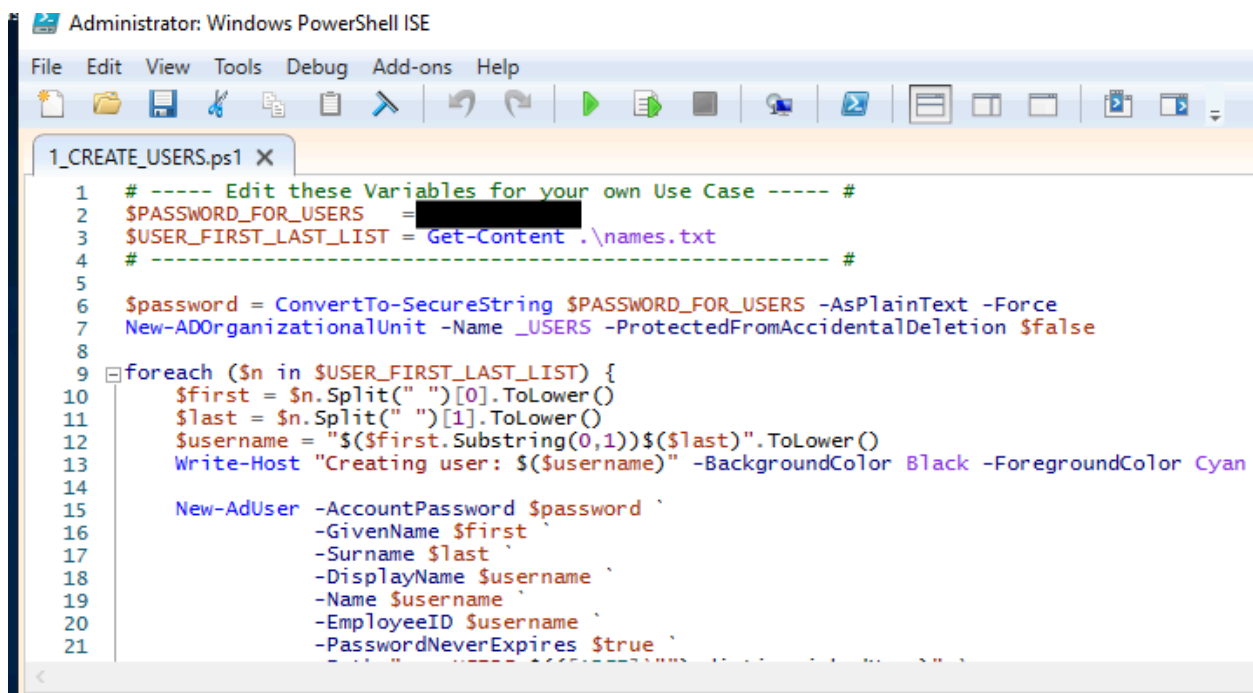
- To see the green arrows (which means our configurations were successful) right click on our server name and click authorize. After, right click on IPv4 and IPv6 and click refresh.

Step 9: Adding users to our active directory via a PowerShell Script.

Source of PowerShell Script: https://github.com/joshmadakor1/AD_PS



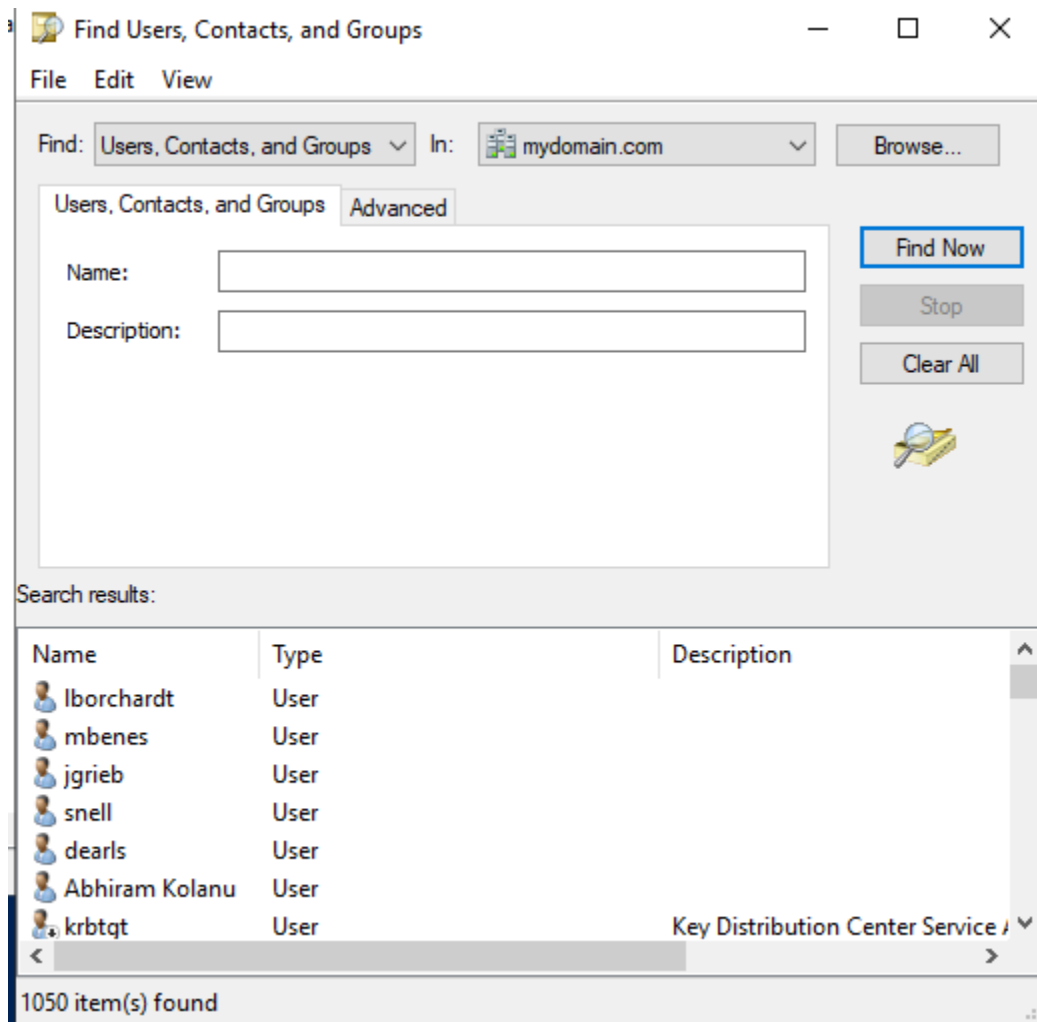
- Once you have downloaded “1_CREATE_USERS.ps1” and “names.txt”, put them in the same directory/folder. Then click start and run “Windows PowerShell ISE” as an administrator.



- Once PowerShell ISE is open, open the script (not names.txt)
- This script will make a user account for all the users in name.txt, so make sure the script and the .txt file are in the same file.

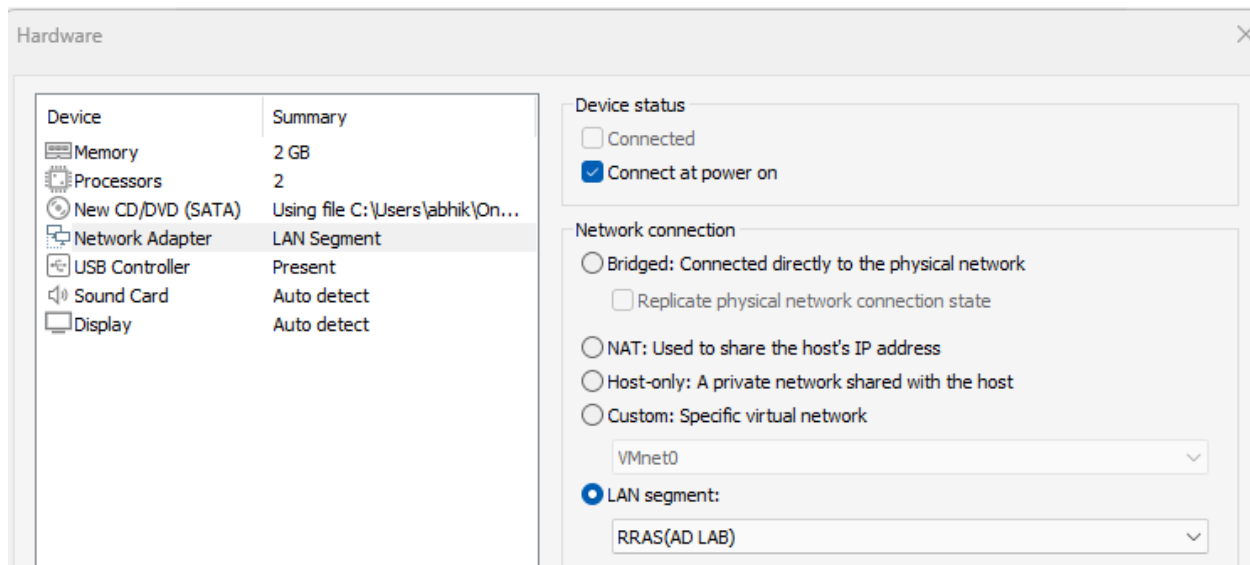
- A simple loop will iterate through the entire file splitting the string into two separate variables, combining the variables into a unanimous format, and then assign the password from the "PASSWORD_FOR_USERS" variable.
- The script will also store all users created by this script into a new organizational unit called "USERS"
- When you run the script make sure to navigate to the directory where you placed the script and the .txt file
 - Ex.

```
PS C:\Windows\system32> cd C:\users\a-abhik\Desktop\AD_PS-master|
```

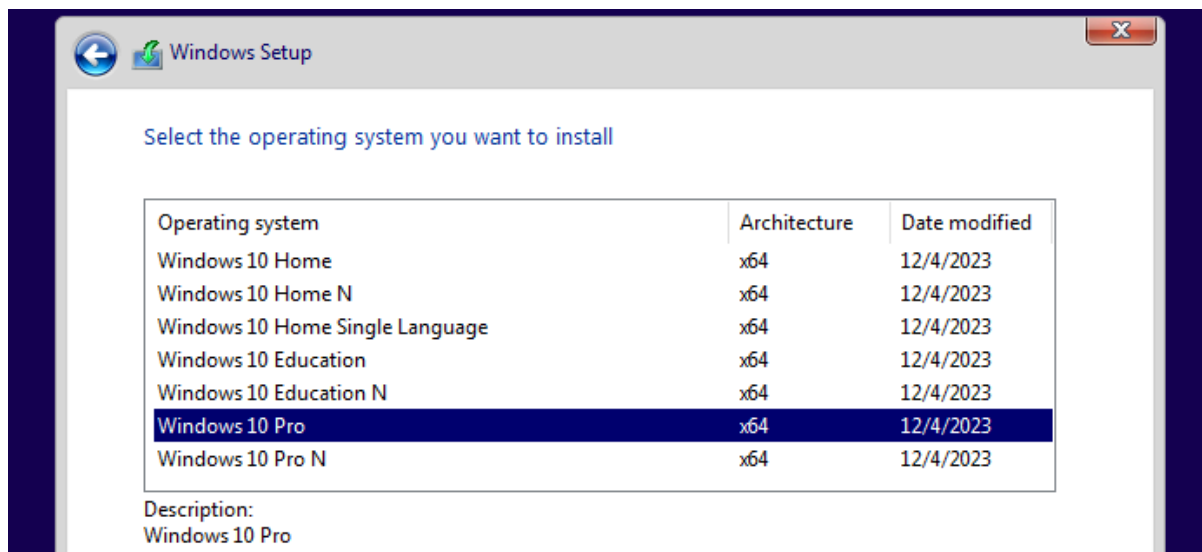


- If we go to "Active Directory Users and Computers", we can see that 1050 users were created and we can search for them by using their username.
- 1000 accounts were created by the script and the other 50 were accounts created upon installation of Active Directory roles.

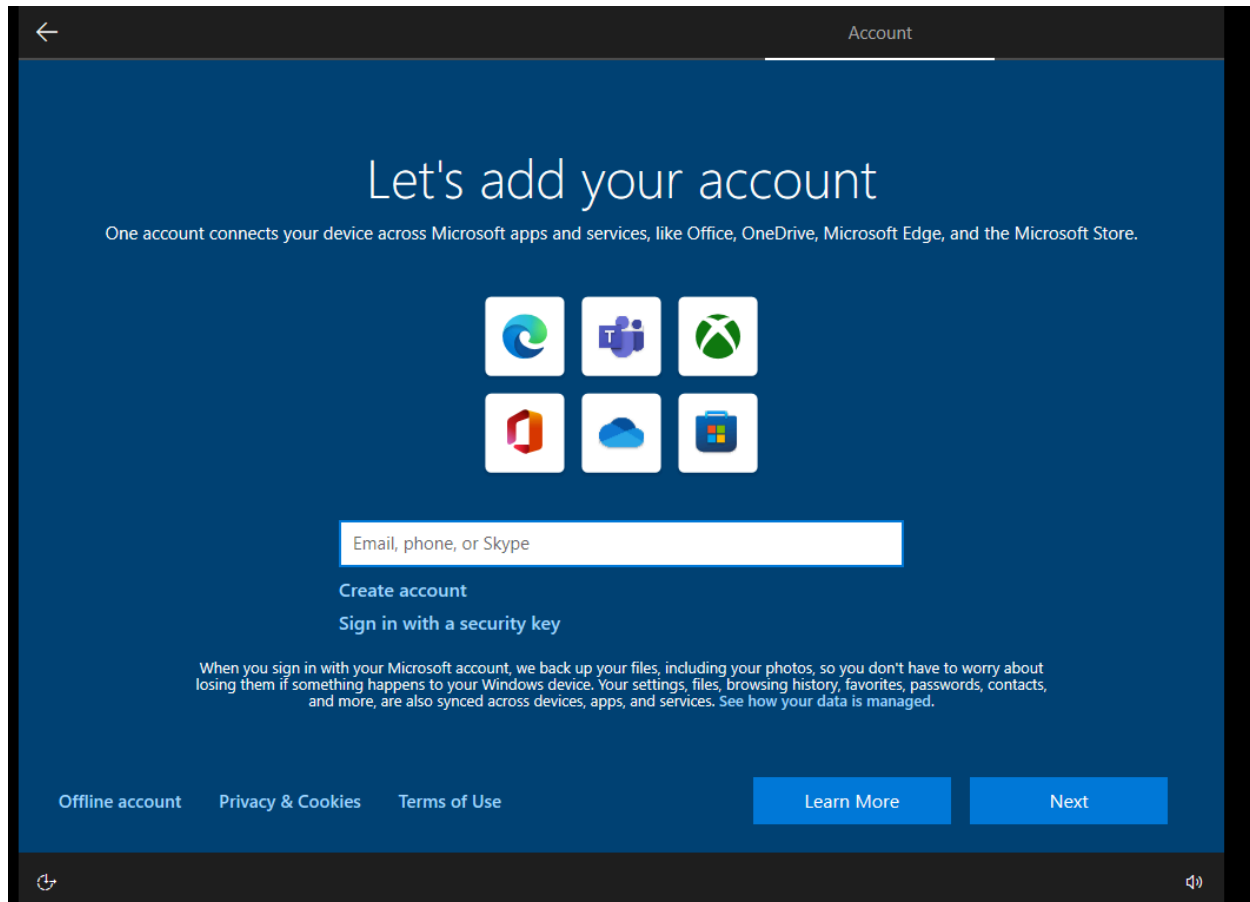
Step 10: Creating Windows 10 Client Machine



- I created a Windows 10 Virtual Machine using a Windows 10 iso file.
- For the Windows 10 Client Machine, we only want it to stay on the internal network. Any communication with the external internet will be enabled and monitored by our Domain Controller. That is why, unlike our DC, the client VM only needs one LAN adapter and the purpose of the LAN Adapter is to connect to the DC using the DHCP scope we configured.



- Make sure to install the Windows 10 Pro operating system since the Windows 10 home causes difficulties when connecting to the DC.



Select Offline account here

- An offline account/Limited Experience is simply an account that is local to the specific machine you install it on
- You can still connect to LAN networks, connect to the DC, and be assigned an IP from the DHCP we configured.
- For the purposes of this lab, an offline account will suffice.

```
PS C:\Users\AbhiramK> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : mydomain.com
    Link-local IPv6 Address . . . . . : 
    IPv4 Address. . . . . : 
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.0.1
PS C:\Users\AbhiramK>
```

If we go to PowerShell and type “ipconfig”, we can see the values that we configured in the DHCP were assigned to our Windows 10 Client VM.

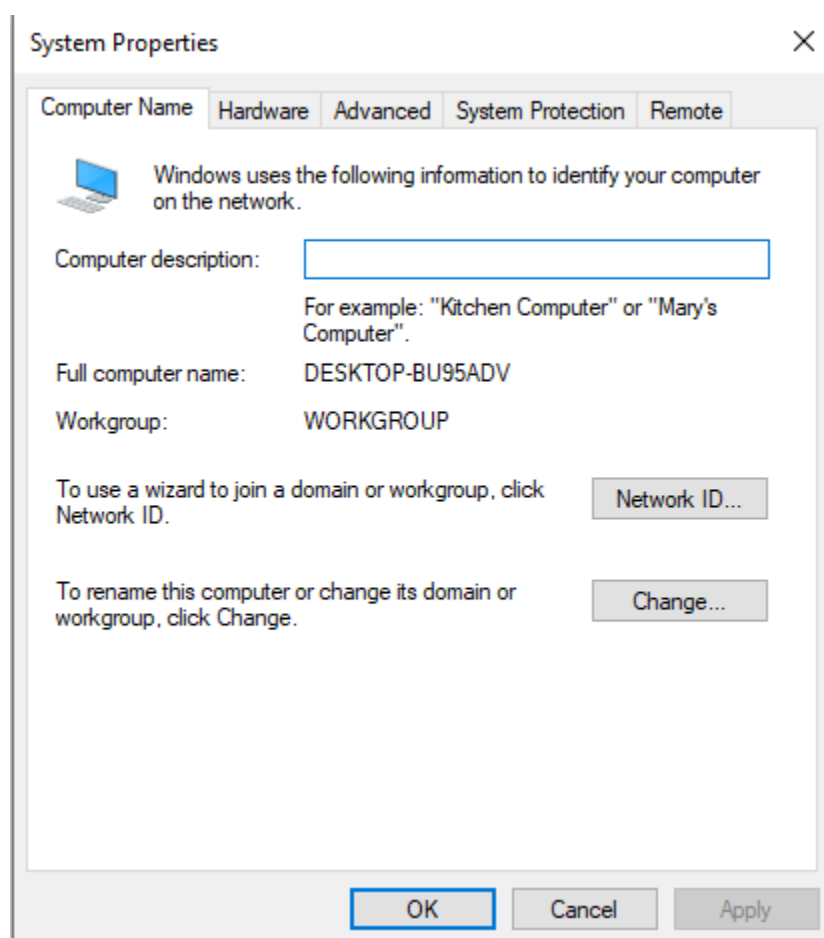
```
PS C:\Users\AbhiramK> ping www.google.com

Pinging www.google.com [142.251.157.119] with 32 bytes of data:
Reply from 142.251.157.119: bytes=32 time=6ms TTL=127
Reply from 142.251.157.119: bytes=32 time=8ms TTL=127
Reply from 142.251.157.119: bytes=32 time=9ms TTL=127
Reply from 142.251.157.119: bytes=32 time=8ms TTL=127

Ping statistics for 142.251.157.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 9ms, Average = 7ms
PS C:\Users\AbhiramK>
```

We were able to ping **www.google.com**, which means that our DNS server is functioning. Successful pings mean that our DNS server is able to translate a human readable name, ex. **www.google.com**, into an IP address that the internet can read.

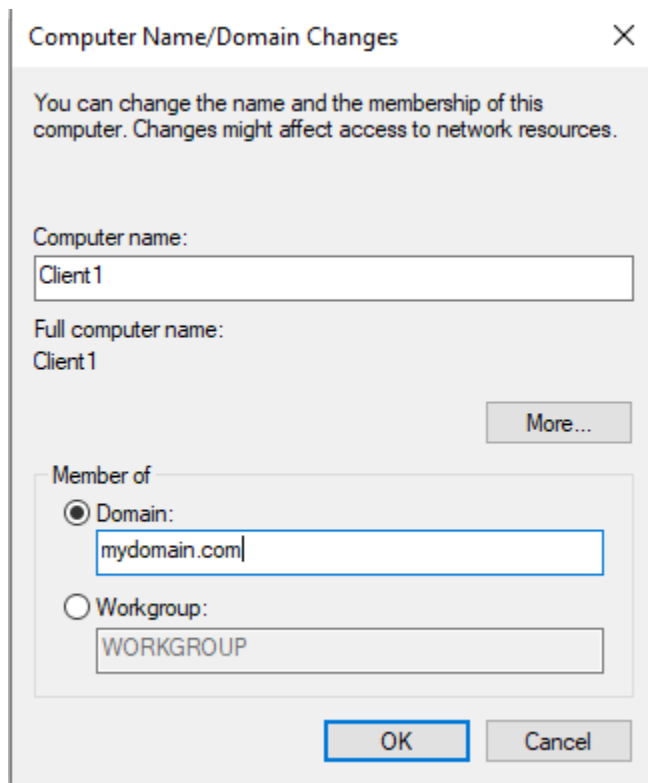
- Pinging means we are getting a response from google, which also means our client VM can access the external internet from a LAN network.



Right click on “Start”, scroll down to “Rename this PC (advanced)” and click “Change...”

- We are doing this to not only rename our PC to a more human-readable name but also connect our client VM to the Domain Controller.

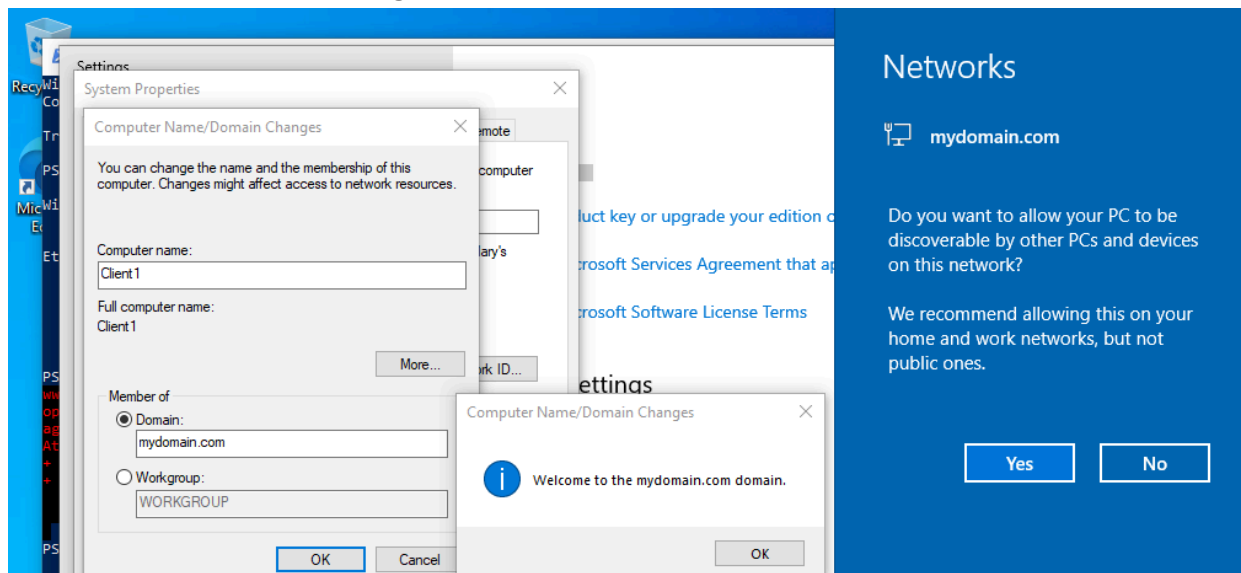
This means we are connecting all the user accounts and Active Directory functions to this client VM, not just the network configurations.

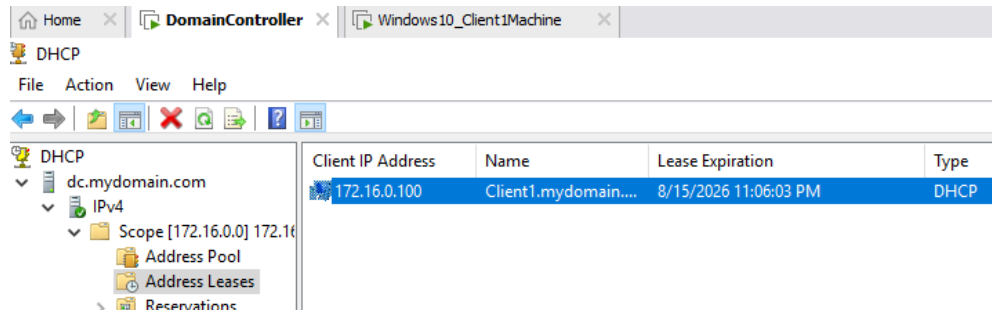


Enter the domain name of the DC under “Domain”.

- You will need to sign in with the admin account we created earlier.

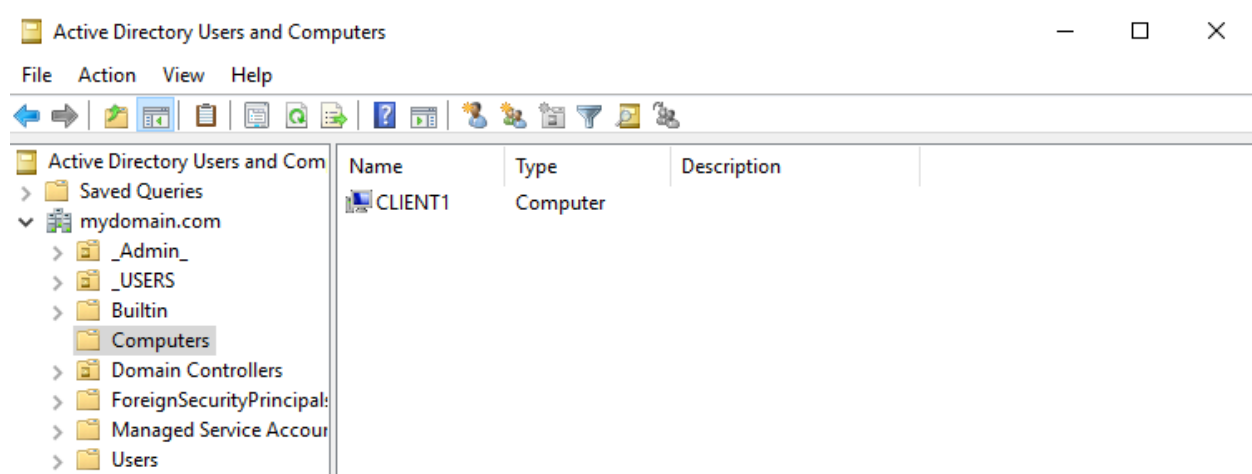
This is what a successful login looks like:





We can also verify the connection was successful by going back to our DC and looking at the DHCP tool.

- We can see here that the client VM we created has a lease of 8 days (this connection was created at 8/7/2026)



Additionally we can also navigate to “Active Directory Users and Computers” and see the name of our computer is listed in the computers folder.

- This means that any of the 1000 accounts we created can be used to login to “CLIENT1”